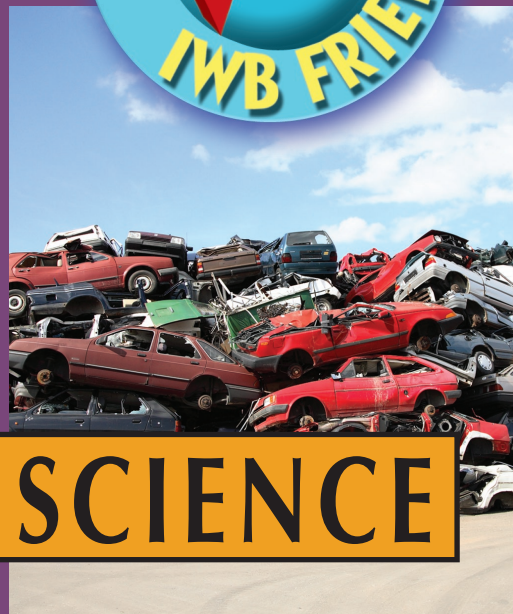


SCIENCE

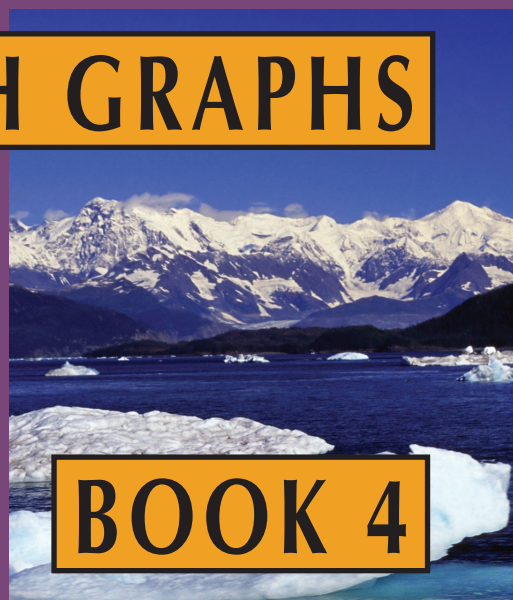
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THE SKILLS OF SCIENCE



DEALING WITH GRAPHS



BOOK 4



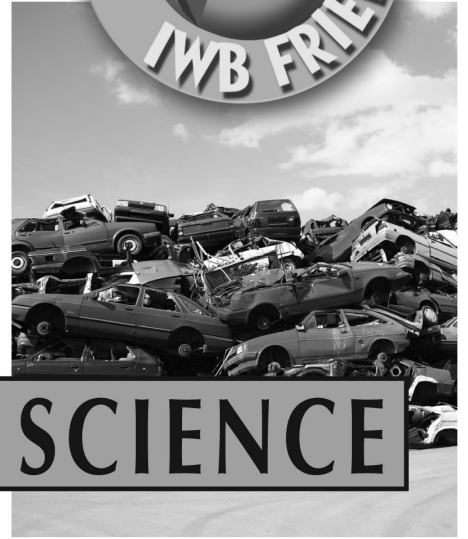
20 sets of PowerPoint slides

Sarah Sharp

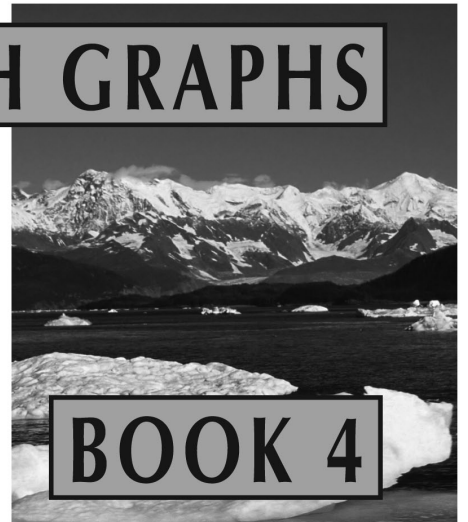


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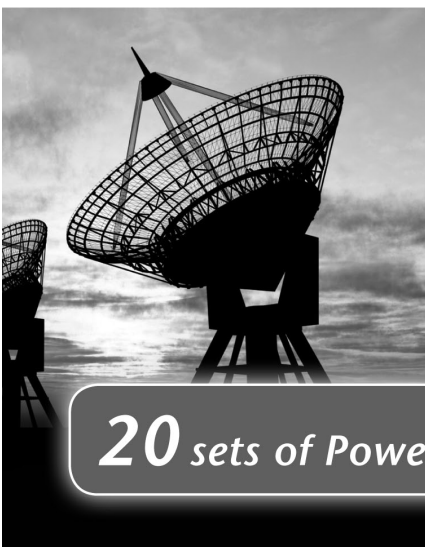


THE SKILLS OF SCIENCE



DEALING WITH GRAPHS

BOOK 4



20 sets of PowerPoint slides

Sarah Sharp

Sarah Sharp has over 10 years' experience teaching Science and Computing Science in comprehensive and selective high schools. She has produced resources for the New South Wales Department of Education and Training on numeracy strategies and the use of dataloggers and their effective integration into teaching and learning programs. Sarah has led teacher workshops in using a variety of computer programs including dataloggers, constructed databases used in recording and analysing student performance and worked with industry providers to critically analyse the design and functionality of software for use in schools. Sarah has been a marker of the Essential Secondary Science Assessment (ESSA) and the School Certificate Science examination papers for several years and is passionate about educating people in science and implementing technology to enrich learning opportunities.

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The *Skills of Science* series is about the nature of science and testing ideas. It encourages an everyday perception of science, conceptual understanding and thinking skills, focusing on how students can learn to think and explore scientifically.

For each title in the series, there is an accompanying CD-ROM containing 20 PowerPoint presentations to summarise and extend the content explored in the topics in the book. These presentations are further complemented by a follow-up activity sheet based on each one, also on the CD-ROM.

The first two books of this series focus on the Scientific Method, and how and why it is used. The third book, *Working with data*, focuses on the collection, description and organisation of data obtained from experiments

This fourth book of the series, *Dealing with graphs*, develops students' skills in graphing organised data to obtain meaning from their results. Graphs are a powerful visual representation of data but they can be misleading if the type of graph is poorly chosen or the graph is incorrectly constructed. There are also specific skills required in order to critically analyse the information a graph contains.

It is divided into three content sections:

- **Graphing skills** introduces students to different types of data and explains how these types are categorised. This leads into different types of graph, and how the data affects the graph type used.
- **Hand-drawn graphs** provides students with the skills they need to choose appropriate graphs for presenting particular data, and to construct them. They will also practise interpreting information from graphs.
- **Spreadsheet graphs** guides students through the process of constructing graphs using Microsoft Excel. Sample data is provided for the initial tasks, so that students can then apply what they have learned to data collected from other sources. Students are also introduced to other functions of Excel that are useful for processing scientific data.

The fifth book, *Science writing made easy*, is based on the premise that important scientific discoveries are more valuable if they are communicated effectively to the wider world. Through the activities, students will explore the effective synthesis and communication of the knowledge and understanding gained from experiments.

Topic	Title	Description	Page
Graphing skills			
1	Types of data	Recognising data as quantitative or categorical	6
2	Types of graphs	Matching the type of data to the most appropriate type of graph	7
3	Drawing a grid scale	Drawing scales for graphs	9
4	Quick quiz	Questions from topics 1–3	10
Hand-drawn graphs			
5	Pie graphs	Drawing pie graphs	11
6	Bar graphs	Drawing bar graphs	12
7	Column graphs and histograms	Drawing column graphs and histograms	13
8	Line and scatter graphs	Drawing line and scatter graphs	15
9	Lines of best fit	Drawing an appropriate line of best fit	17
10	Extrapolating and interpolating	Extrapolating and interpolating from line and scatter graphs	18
11	Linear relationships	Determining linear relationships from scatter graphs – Ohm’s Law	19
12	Non-linear relationships	Determining non-linear relationships from scatter graphs – The Wave Equation	21
13	Composite pie and bar graphs	Interpreting composite pie and bar graphs	23
14	Divided bar graphs	Drawing and interpreting divided bar graphs	24
15	Composite line and divided bar graphs	Drawing composite line and divided bar graphs	26
16	Quick quiz	Questions from topics 5–15	28
Spreadsheet graphs			
<i>– using Microsoft Excel 2003/2007</i>			
17	Excelling at pie graphs	Constructing pie graphs, SUM function, inserting rows	29
18	Excelling at bar graphs	Constructing bar graphs, understanding and copying simple formulas, SUM and AVERAGE functions	31
19	Excelling at column graphs	Constructing column graphs, making superscripts and subscripts	33
20	Excelling at histograms	Constructing histograms, creating a series	35
21	Excelling at line graphs	Constructing line graphs, entering scientific notation, adjusting decimal places and scales	37
22	Excelling at scatter graphs	Constructing scatter graphs, selecting non-adjacent columns	39
23	Excelling at divided bar graphs	Constructing divided bar graphs, merging and text wrapping cells	41
24	Excelling at composite graphs A	Constructing composite graphs for multiple sets of data for one y-axis variable, printing worksheets and revision of previous skills	43
25	Excelling at composite graphs B	Constructing composite graphs for two y-axis variables, widening rows and revision of previous skills	45
Selected answers			47

What does it include?

The topics in this title are complemented by the information and activities contained on the CD-ROM at the back of this book.

The CD-ROM includes:

- 20 PowerPoint presentations to accompany the topics in this book, which summarise and extend the content.
- 20 follow-up activity sheets, each based on the matching PowerPoint presentation. These are supplied as PDF documents.

Note to the teacher

To get the most out of these teaching resources, teachers are encouraged to preview the PowerPoint presentations and accompanying activity sheets before using them in class.

Using the CD-ROM

In the Windows environment, the CD-ROM will autorun. From the Main menu, simply click the PowerPoint presentation you wish to view.

In the Macintosh environment, double-click the 'Science – Dealing with graphs' icon.

From the Main menu, click the PowerPoint presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

The Slide Show view is used to display each PowerPoint presentation on the screen.

It is recommended that you use an external projector for whole-class use.

To access the activity sheets, simply click the pencil icon on the appropriate topic on the main menu.

Technical specifications

Windows

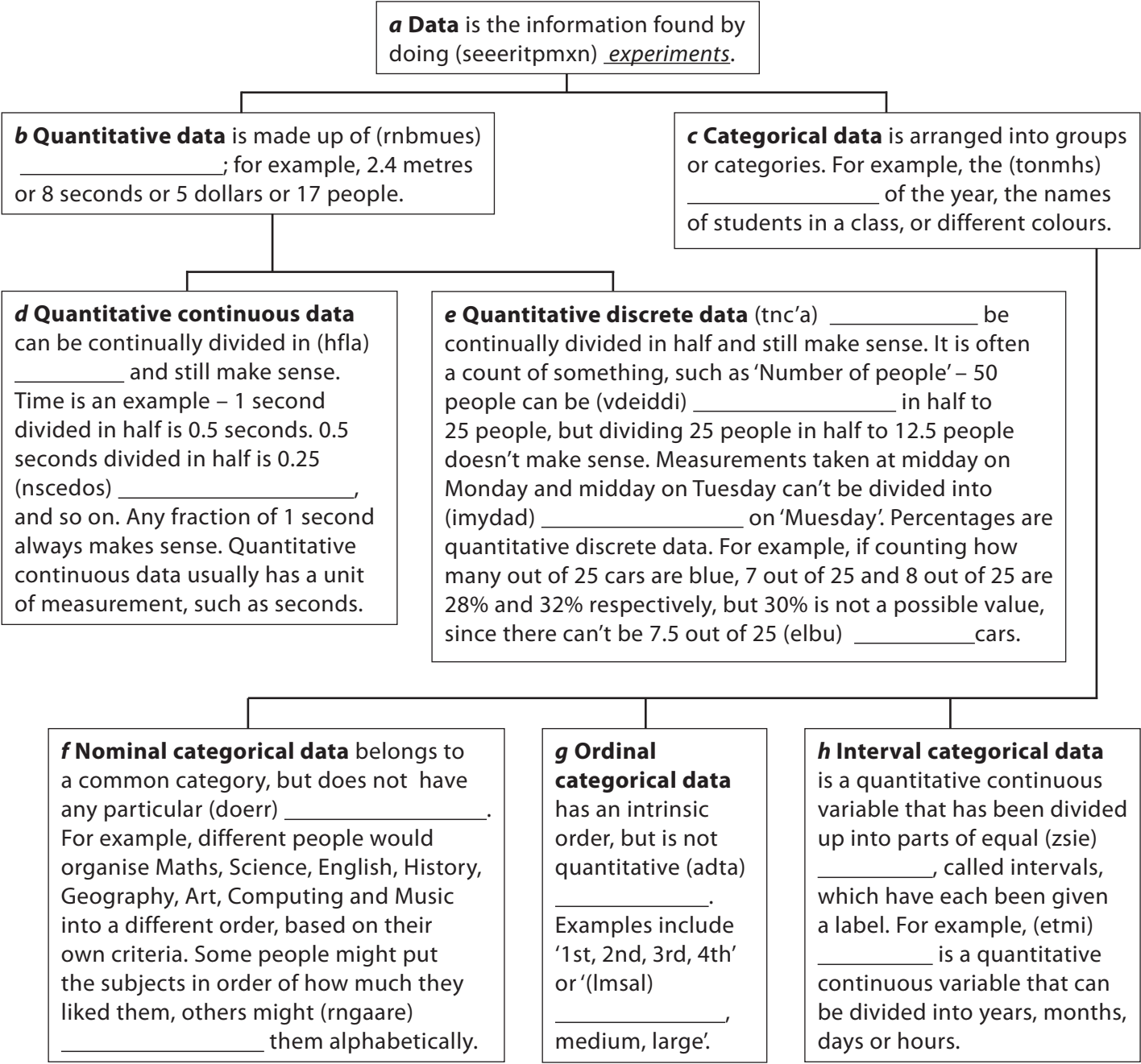
- The CD-ROM will autorun in the Windows environment, with an operating system of Windows XP or later.
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a version prior to Office 2000, some cosmetic losses may occur. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Macintosh

- Double-click the 'Science – Working with Data' icon on the CD-ROM. From the Main menu, click on the presentation you wish to view.
- To view a fully featured presentation, and edit it, you must have the PowerPoint 2004 application installed (i.e. Office 2004 for Macintosh or later).
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a Macintosh environment, some cosmetic losses may occur. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Data is the information found by doing experiments and it can be divided into several different types. Data on its own is not very useful – it needs to be organised into tables and processed into graphs. Knowing the type of data helps you to decide which type of graph is the most appropriate to use to display the data.

1 Unjumble the words in brackets to complete the descriptions of the different types of data. The first one has been done for you.



2 Draw lines to match each type of data with its example.

TYPE OF DATA	EXAMPLE
• quantitative discrete	a The grades 'poor, satisfactory, average, above average, excellent' on a report card
• interval categorical	b The number of crabs in a rockpool at 12 pm each day
• quantitative continuous	c The colours of Smarties in a bag
• nominal categorical	d The distance between Sydney and Melbourne
• ordinal categorical	e The weight classes 'underweight, healthy weight, overweight, obese'



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Graphs are an important tool for communicating data. Different types of graphs are used to show different types of data.

1 Complete the cut and paste activity on the next page. You will need scissors and glue.

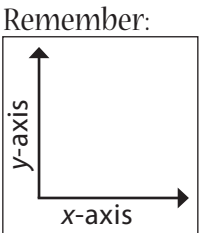
The graphing descriptions and examples in the cut and paste activity you did can be used to find the most appropriate graph for each type of data. You will often see graphs where the type of data does not match the type of graph. For example, joining the plotted crosses in a line graph suggests that the points on the line between the crosses have meaning. But line graphs are sometimes used to show trends in categorical data when there is no meaning to the points between the crosses. Also, the different types of categorical data graphs are often used interchangeably.

Use the information from the cut and paste activity to complete the following questions.

- 2 Several different types of graphs are used to display categorical data. List all of them.
- 3 Which graph is the most appropriate to use to show each type of data?
- a The data involves a group of widely different percentages. _____
 - b The data consists of frequencies. _____
 - c The x-axis data is categorical data and the y-axis data is quantitative continuous data. _____
 - d Finding a relationship between two quantitative continuous variables. _____

The x-axis data is listed in the first column in a data table.
The y-axis data is found in the second column of the data table. However:

- for bar graphs the y-axis data is in the first column and the x-axis in the second column
- pie graphs have no axes.



- 4 Identify the most appropriate graph to represent the data shown in each table.
Hint: First identify the type of data which would be on the x-axis and y-axis.

a _____

MAJOR PLANET	DIAMETER (km)
Mercury	4 900
Venus	12 100
Earth	12 800
Mars	6 800
Jupiter	143 000
Saturn	120 000
Uranus	51 800
Neptune	49 500

b _____

HOUSEHOLD ENERGY USE	PERCENTAGE (%)
Room heating	47
Lighting and appliances	24
Water heating	17
Air conditioning	6
Refrigerator	5

- 5 Write the column heading for the data that would be on the x-axis for the tables in questions 4a and 4b.
- 4a _____
 - 4b _____
- 6 Identify which graph is the most appropriate to draw in the following situations.
Hint: First identify the type of data which would be used on the x-axis and y-axis.
- a The relationship between the length of the string on a pendulum and the average time it took to make one swing was investigated. _____
 - b The total number of salmon swimming upstream was counted every month.

 - c The velocity of a car was measured every minute for 30 minutes. _____

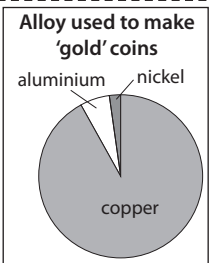


- 1 a Write the heading 'Graphing Data' in your workbook. Read the information in the box below.
Then cut out the box and paste it under the heading in your workbook.
- b Now cut out and match each graph with the correct description. *Hint:* Check with your teacher that your matches are correct **before** pasting them into your workbook.

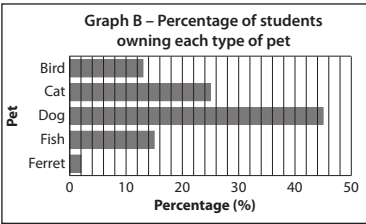
Types of data – definitions

- Quantitative data is made up of numbers.
- Quantitative continuous data can be continually divided in half and still make sense. Example: measurements of length and time.
 - Quantitative discrete data cannot be continually divided in half and still make sense. Example: a count of the number of cars in a parking lot.
- Categorical data is arranged into groups or categories.
- Categorical nominal data has no order to the categories. Example: red, green, yellow, purple
 - Categorical ordinal data has an order to the categories. Example: 1st, 2nd, 3rd
 - Categorical interval data is quantitative continuous data divided up. Example: time is divided into January, February, March, and so on.

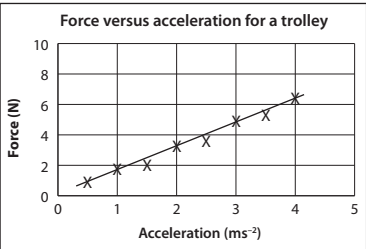
Pie graph



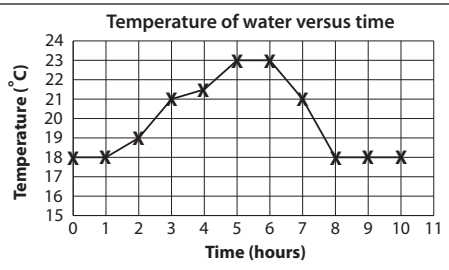
Bar graph



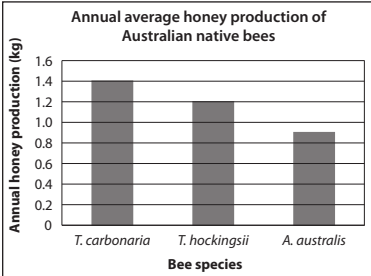
Scatter graph



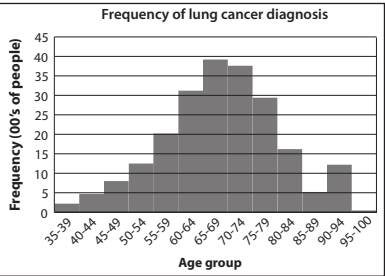
Line graph



Column graph



Histogram



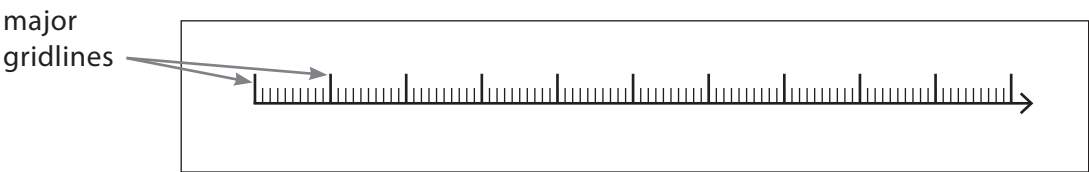
- Used when the dependent variable is a count of how often something occurs, also known as a *frequency*.
 - Example: Showing the number of people diagnosed with lung cancer in each age group
 - x-axis data: Categorical ordinal or interval
 - y-axis data: Quantitative discrete
- Used to show a set of percentages or fractions that have large differences in the size of the proportions. Graph has no x- or y-axes.
 - Example: Showing the amount of different metals in an alloy.
 - If a bar graph was used to plot the data:
 - x-axis data: Quantitative discrete
 - y-axis data: Categorical nominal or ordinal
- Used to show a **trend** between unrelated variables, usually over time.
 - Example: Showing how temperature changes over time.
 - x-axis data: Quantitative continuous
 - y-axis data: Quantitative continuous
- Used to show categorical data with a quantitative continuous y-axis variable.
 - Example: Showing the mass of honey produced annually by different bee species.
 - x-axis data: Categorical
 - y-axis data: Quantitative continuous
- Used to show a set of percentages or fractions that have small differences in the size of the proportions.
 - Example: Showing the percentage of students owning different types of pets.
 - x-axis data: Quantitative discrete
 - y-axis data: Categorical nominal or ordinal
- Used to find a **relationship** between an independent and dependent variable.
 - Example: Showing that force increases as acceleration increases.
 - x-axis data: Quantitative continuous
 - y-axis data: Quantitative continuous

Grid scales are used when constructing graphs with axes. There are two types of grid scale – quantitative and categorical. Quantitative data is made up of numbers. Categorical data is divided into groups, or categories.

Sometimes it's easiest to use trial and error. Try counting along the grid, increasing by 1 or 10 or 25 or 50 or 100 or another whole number at each major gridline, so that all the grid space is used.

Quantitative data grid scales

- Always start with zero.
 - Increase by the same amount each division.
- 1 Follow the steps below to draw a quantitative data scale on the grid for the data: 17, 65, 88, 43.



- Step 1** Count the number of major gridlines along the grid, not including the starting gridline. _____
- Step 2** Round up the largest number to be included in the scale to the nearest 1, 10, 25 or 100. 88 is the largest number in this example and it would round up to 90.
- Step 3** Divide the number in step 2 by the number in step 1. _____ ÷ _____ = _____
- Step 4** Round up the answer from step 3 to the nearest 1, 10, 25 or 100. In this example, 9 would round up to 10. This is how much the scale should increase for each major division.
- Step 5** Mark the scale along the grid. Start at 0 and increase by the answer in Step 4 at each major gridline.

2 Draw grid scales for the following sets of quantitative data.

a 3, 5, 9, 1, 7, 12



b 158, 27, 64, 79, 121



Sometimes there is a big difference between zero and the numbers in a scale. To avoid wasting space, a scale can be broken using either of these symbols:  

A scale should only be broken once, and usually only at the start.

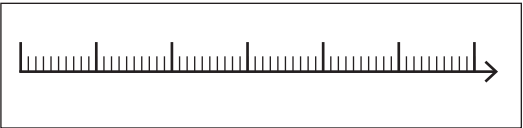
- 3 Draw a broken quantitative data scale for these numbers: 76, 89, 64, 67, 83.
Hint: Using the steps above, in step 1 count from the first major division **after** the break.



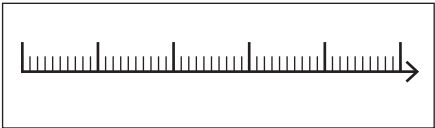
Categorical data grid scales

- Only start with zero if it is one of the categories.
 - Have each category the same width.
 - Are placed in order from smallest to largest, if applicable.
 - Have an equal-sized gap on either side of each category, except in histograms which have no gaps.
- 4 Draw a grid scale for each set of categorical data.

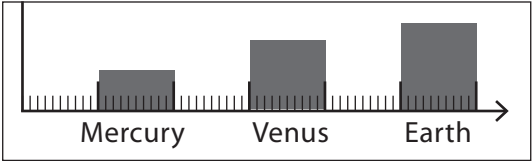
a A, B, C



b 0–9, 20–29, 10–19, 30–39 (histogram)



Example:

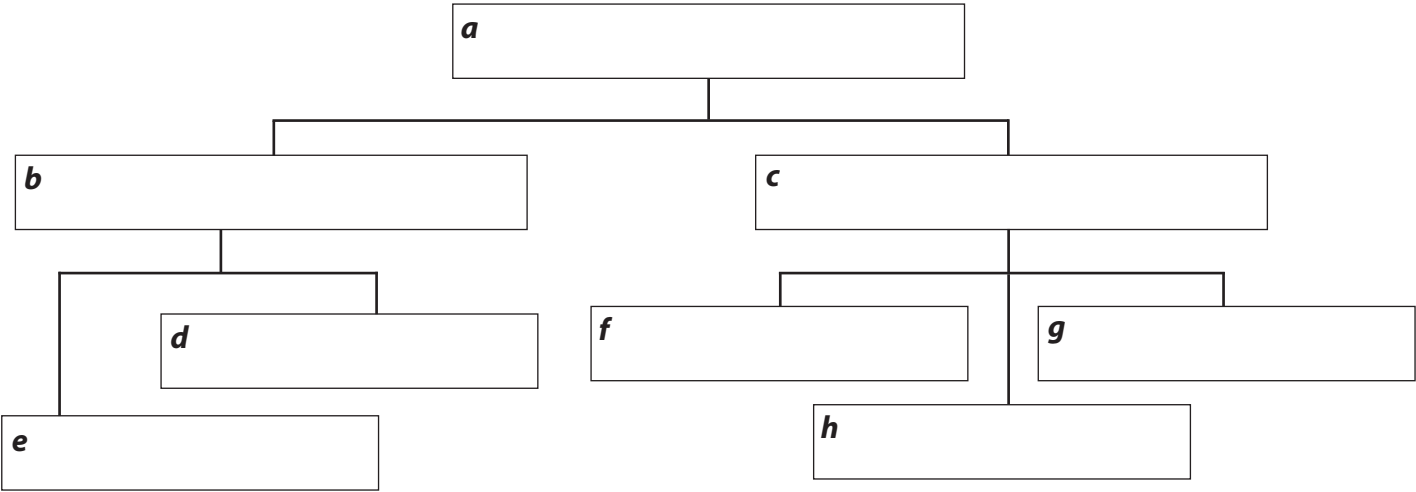




Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

1 Use the word bank to complete the organisational chart on types of data.

WORD BANK	• Discrete	• Quantitative	• Categorical	• Continuous
	• Ordinal	• Data	• Interval	• Nominal



2 Identify the type of data represented by each description.

- a The type of pet students own _____
- b The number of pets each student has _____
- c The age ranges 10–19, 20–29, 30–39, 40–49 _____
- d The mass of different students _____

3 Write the most appropriate graph to use to display the set of data in each description.
Graphs: *line, histogram, column, pie/bar, scatter.*

DESCRIPTION	GRAPH
a Shows the mass of each planet in our solar system	
b Shows the relationship between the mass of a species of animal and its brain mass	
c Shows the proportion of elements present in the human body	
d Shows the velocity of a motorbike each minute during a 15 minute journey	
e Shows the number of students within different 5 cm height ranges in a class	

4 Often several types of graphs could be used to plot a set of data, even if only one type is the most appropriate.

- a Identify the types of graphs that could be used to plot the information in the table.

- b Which of the graphs you identified would be the most appropriate?

LOCATION OF WATER USE	PERCENTAGE PER HOUSEHOLD
Bathroom	20
Toilet	16
Laundry	13
Kitchen	8
Outdoors	43

5 On the grid at the right, draw the scale that would be used to plot this quantitative data:

75, 68, 82, 70, 84.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

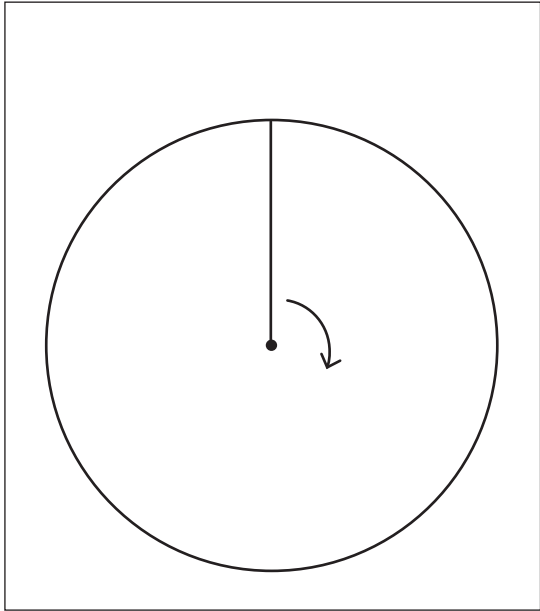
Pie graphs are used to show each of the parts that make up a whole. They are an appealing way of quickly communicating which percentages are the biggest or smallest, and a useful type of graph for showing large differences in the size of proportions.

You need a protractor to complete this activity.

- 1 Follow the steps to draw a pie graph showing each of the ways students travel to school.

Step 1 Put the data into a table. The data recorded was
Bus – 42%, Walk – 8%, Car 30%, Train – 17%, Bike – 3%.

METHOD OF TRAVEL	PERCENTAGE OF STUDENTS	ANGLE



Step 2 Pie graphs are in the shape of a circle. All circles are made up of 360°. To work out how many degrees are represented by a method of travel, multiply its fraction by 360°. If the data is written as a percentage, write it as a fraction by putting it over 100.

For example: Bus proportion = $\frac{42}{100} \times 360^\circ = 151.2^\circ$
There is no need for decimal places when drawing pie graphs, so round 151.2° to the nearest whole degree, 151°, and write it into the appropriate place in the third column of the table.
Repeat this step for the four other methods of travel.

Step 3 Write a title above the circle at the top right of this page. The title should tell the reader exactly what data the graph represents. For example: The different ways students travel to school.

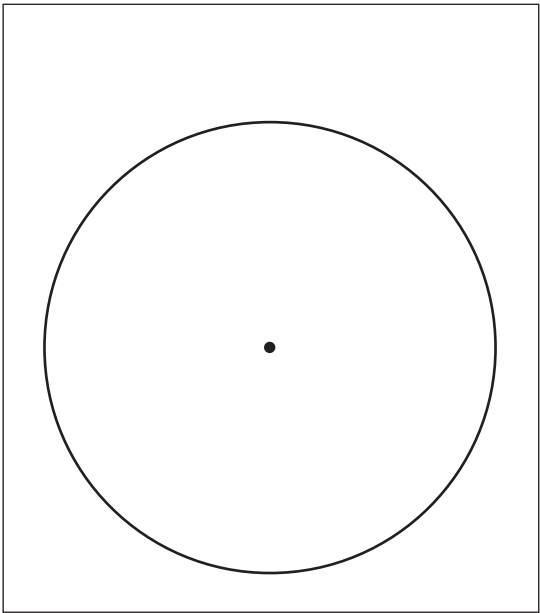
Step 4 Underneath the title, a compass has been used to draw a circle with a radius of at least 3 cm. A vertical line has been drawn from the top circumference of the circle to the centre to start the pie graph.

Step 5 Use a protractor to measure the first angle, 151°, from the vertical line in a clockwise direction. Mark the angle in pencil on the circumference. Then draw a line joining the mark to the centre of the circle to form the first pie piece. Continue measuring angles from the previous line until all methods of travel are included.

Step 6 Each piece of pie graph is called a sector. Label each sector with the method of travel and its percentage.

- 2 Draw a pie graph for the following data on the species of fish caught in 2007 from a local lake.

FISH SPECIES	PERCENTAGE CAUGHT	ANGLE
Flathead	72	
Perch	14	
Rainbow fish	9	
Other	5	



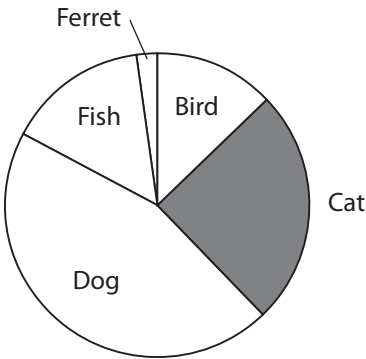
Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

The size of the sectors of a pie graph can be difficult to estimate accurately, especially if there isn't a big difference in the size of the proportions. Bar graphs allow the proportions to be compared more easily.

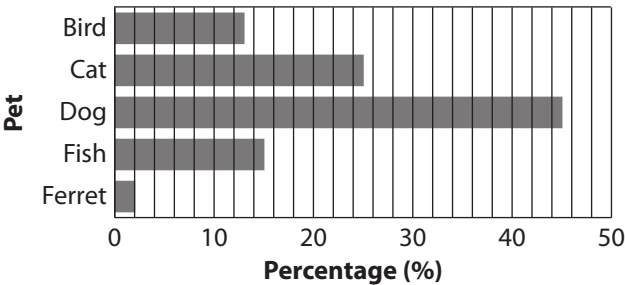
1 Estimate the percentage of students who own cats in:

- a pie graph A _____ b bar graph B _____.

Graph A – Pets owned by students



Graph B – Percentage of students owning each type of pet



Bar graphs are the only type of graph to have the first column in a data table plotted on the y-axis.

Both graph A and graph B show that 25% of students own cats, but it is much easier to see this in graph B.

2 In graph B:

- a which is the least common pet? _____
b what percentage of students owned a pet that is a mammal? _____
c If 200 students were surveyed, how many students had pet fish? _____

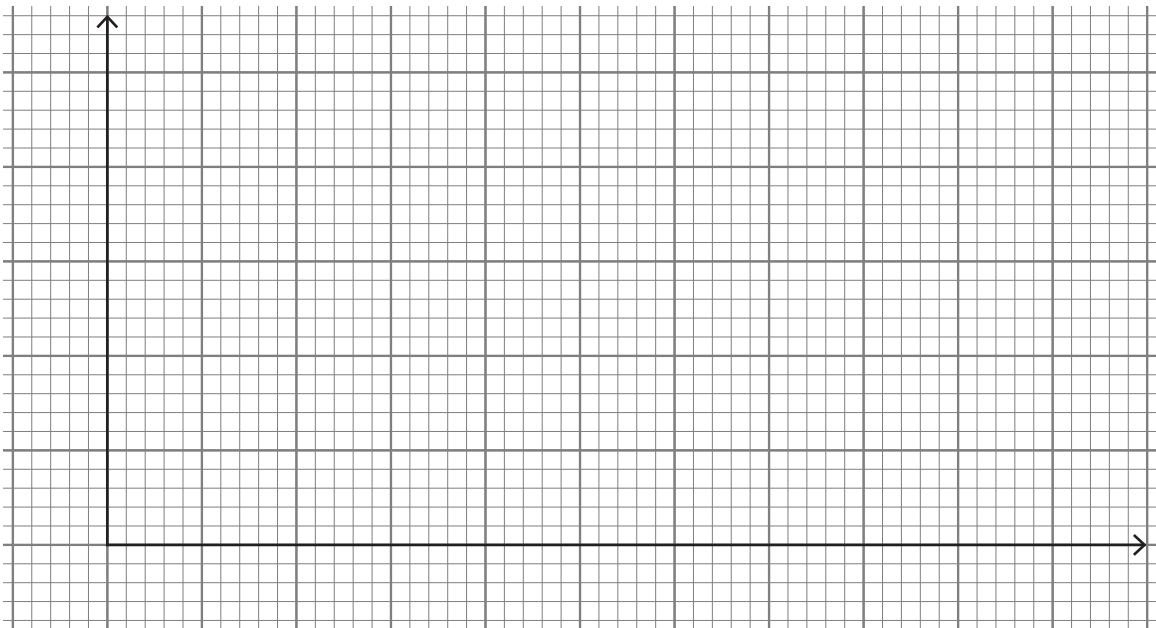
3 a Complete the table to find the percentage of each ingredient in an apple pie. Round the percentages off to whole numbers.

Hint: Add all the masses to find the total mass, then find the percentage for each ingredient using the formula:

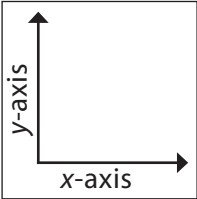
$$\frac{\text{ingredient mass}}{\text{total mass}} \times 100 = \text{percentage mass of ingredient}$$

- b Draw a bar graph to show the percentage of each ingredient in apple pie. Hint: Use graph B above as an example of all the necessary parts to include when drawing a bar graph.

INGREDIENT	MASS (g)	PERCENTAGE
Pastry	95	
Sugar	15	
Apple	80	
Butter	20	
TOTAL		



Remember:



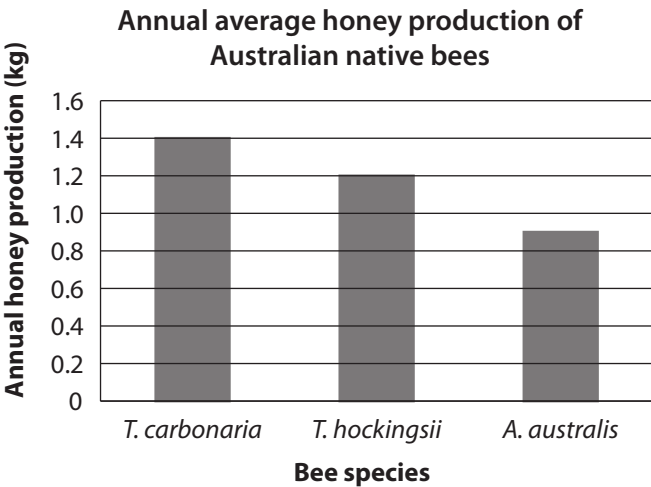
Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

- Although they look very similar, column graphs and histograms display different types of data.
- **Histograms** show a frequency distribution of categorical data, which is a count of how often something occurs for each category.
 - **Column graphs** also show categorical data on the x -axis, but the data on the y -axis is quantitative continuous, which is a measurement of something rather than a count of how often it occurs.

Data on the average annual honey production from one hive of several different Australian native bee species is shown in the table at the right. The data has been represented by a correctly drawn column graph.

BEE SPECIES	AVERAGE HONEY PRODUCED PER HIVE (kg)
T. carbonaria	1.4
T. hockingsii	1.2
A. australis	0.9

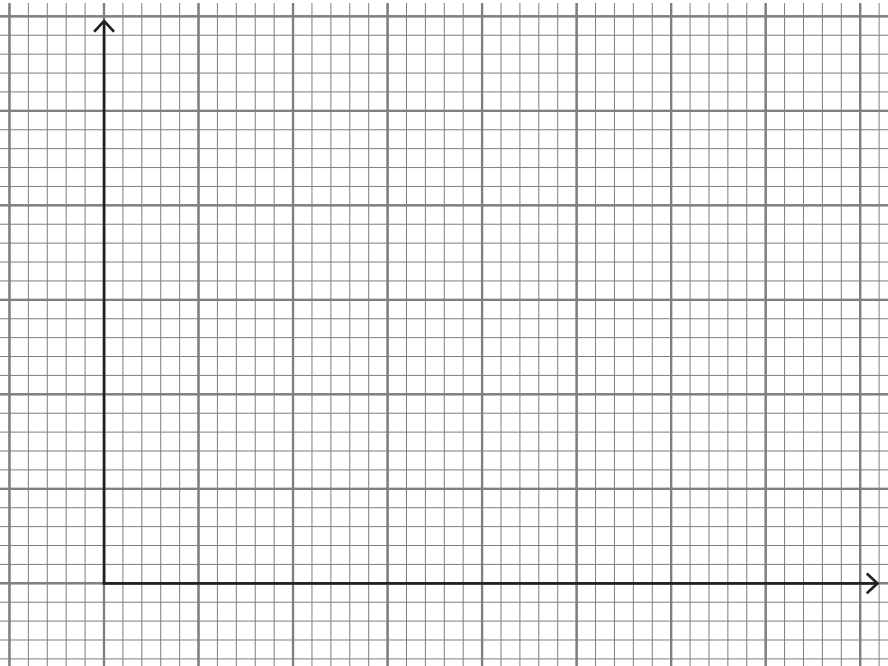
- 1
- List at least five features of a correctly drawn column graph.
-



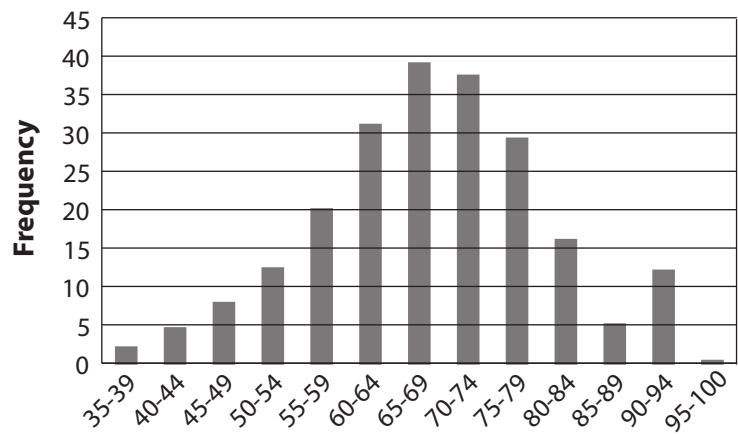
- 2 a
- Draw a column graph on the grid provided to show the average adult length of different species of whales. *Hint:* Use the list you completed to make sure you include necessary parts of the graph.

WHALE SPECIES	AVERAGE ADULT LENGTH (m)
Humpback	15
Orca	9
Southern Right	17
Blue	27

- b
- Rank the whales from longest to shortest.
-



- 3 Column graphs and histograms are drawn the same way, except column graphs have equal-sized gaps between the columns and histograms have no gaps between the columns.
- a A histogram was drawn showing the frequency of lung cancer diagnoses in Canada in 2006. Circle at least five mistakes that were made in drawing this histogram, and add a label to briefly describe each mistake.

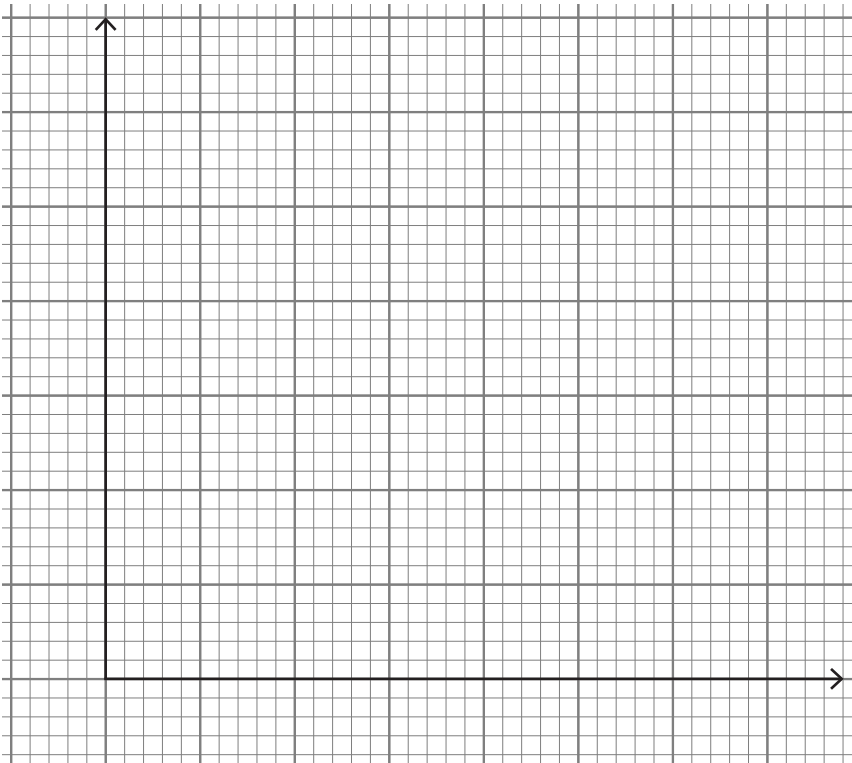


AGE GROUP (YEARS)	FREQUENCY (00'S OF PEOPLE)
35–39	2
40–44	4.5
45–49	7.8
50–54	12.3
55–59	20
60–64	31
65–69	39
70–74	37.4
75–79	29.2
80–84	16
85–89	5
90–94	2
95–100	0.25

- b At what age was a Canadian most likely to be diagnosed with lung cancer in 2006?
- c Propose one reason why the age ranges start at the 35–39 age group.
- d Write the two age groups that would come before the 35–39 age group.

- 4 a Use the information provided in the table to draw a histogram of the number of eggs laid per nest for a population of doves in 2006.

EGGS PER NEST	FREQUENCY
1	9
2	33
3	27
4	10
5	2
6	1



- b What is the most frequent number of eggs laid in doves' nests? _____
- c What is the least frequent number of eggs laid in doves' nests? _____
- d How many dove nests were sampled? _____

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Line graphs show trends in data, usually over a period of time. The trend may go up or down or both, but the *y*-axis variable is not dependent upon the *x*-axis variable.

For example, the graph below shows how the heart rate of a person changes as they move from sitting on a chair, to jogging on the spot for 5 minutes then to sitting on the chair again. Their heart rate is not affected by what time it is, but by their amount of physical activity. The trend shown in the graph would probably be different if it was measured over a different 10 minute period or for a different type of physical activity.

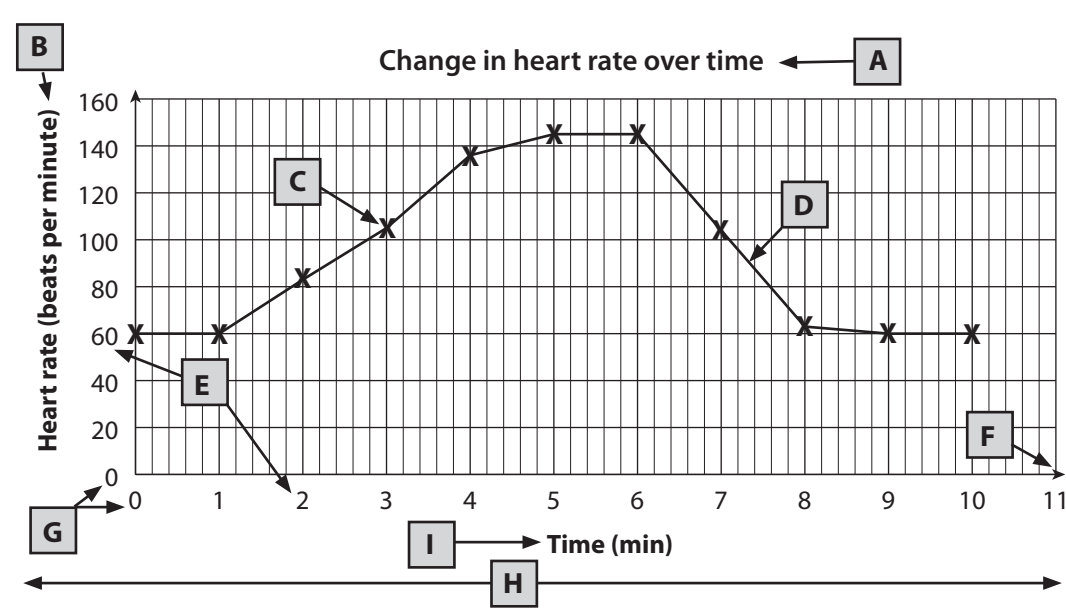
Scatter graphs are used to show the relationship between an independent variable and a dependent variable. Measurements of the force on an object and its resulting acceleration would be expected to show the same relationship, no matter when or where or how the experiment was performed. The acceleration of an object always depends upon the force applied to it.

Line and scatter graphs are drawn in exactly the same way, except that for a line graph, the points are joined dot-to-dot and for a scatter graph, a line of best fit is drawn.

In any experiment, there is one set of data where the values are chosen – this is the *independent variable*. The corresponding set of data that is measured is the *dependent variable*.

The *line of best fit* on a scatter graph shows the relationship between the independent and dependent variables. It should have an equal number of points on either side of the line, if some points do not fall exactly on the line.

1 Identify the necessary parts of a line graph by writing the letters on the graph beside the correct feature.



FEATURES	
	evenly spaced scales
	graph title
	units for axis title
	axis title
	ruled line joining each point
	graph takes up maximum space
	'x' for each plotted point
	scales start at 0
	ends of axes have arrowheads

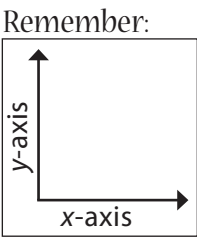
2 Refer to the graph to answer the following questions.

- a Over what period of time was the heart rate measured?

- b At what time was the maximum heart rate reached?

- c During what time interval(s) was the heart rate constant?

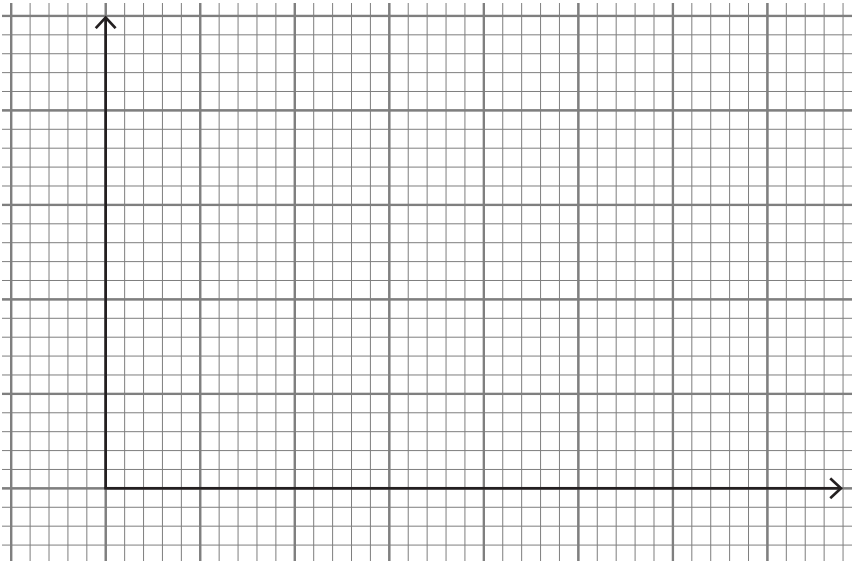
- d What are the units of measurement of the data on the y-axis?



In any experiment, there is one set of data where the values are deliberately changed – this is the independent variable. It is written in the first column of a data table and it is plotted on the *x*-axis. The corresponding set of data that is measured is the dependent variable. It is written in the second column of a data table and is plotted on the *y*-axis.

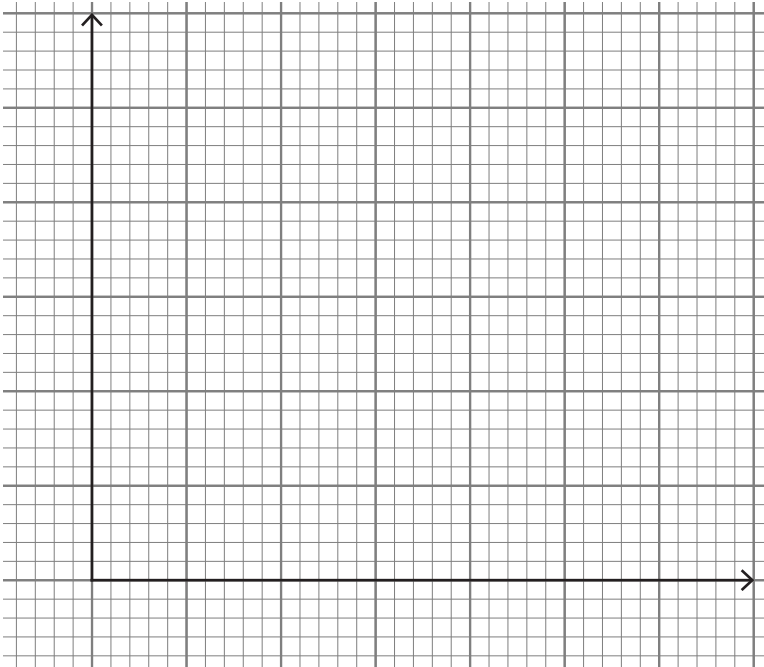
3 On the grid below, draw a line graph of the data on the velocity of a car at different times. Be sure to include all the features you identified in question 1.

TIME (min)	VELOCITY (ms ⁻¹)
0	14
1	14
2	14
3	0
4	23
5	32
6	29
7	36
8	19
9	0



4 a Draw a scatter graph to show how the force needed to pull a trolley changes with its mass.
Hint: Scatter graphs are drawn the same way as line graphs, except a line of best fit is drawn rather than joining the points dot-to-dot.

MASS (g)	FORCE (N)
50	2.4
100	5.1
150	7.8
200	9.5
250	12.3
300	15.0
350	16.9

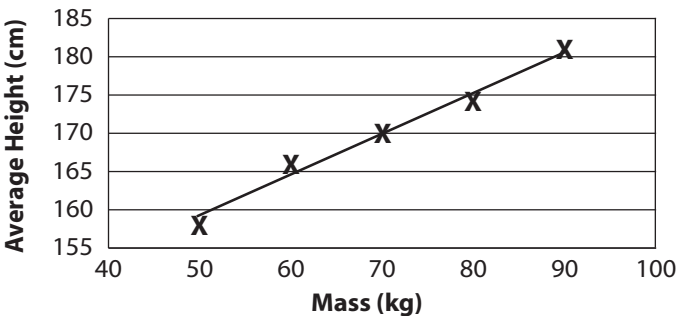
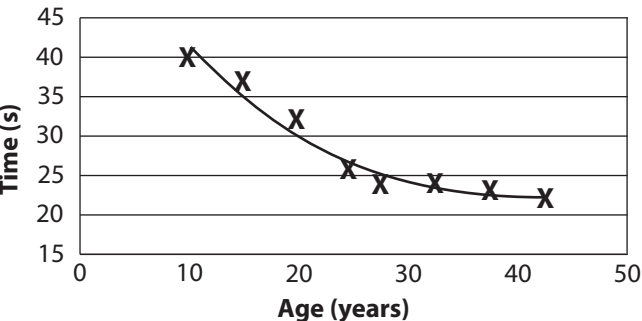


b Describe the relationship between force and mass shown in the graph.

5 A good title for scatter graphs is 'y-axis title versus x-axis title'. Write a title for each graph below.

a Title: _____

b Title: _____



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



The line of best fit on a scatter graph shows the relationship between the independent and dependent variables. The independent variable is plotted on the *x*-axis and the dependent variable is plotted on the *y*-axis.

1 Use the word bank to complete the rules for drawing lines of best fit.

WORD BANK

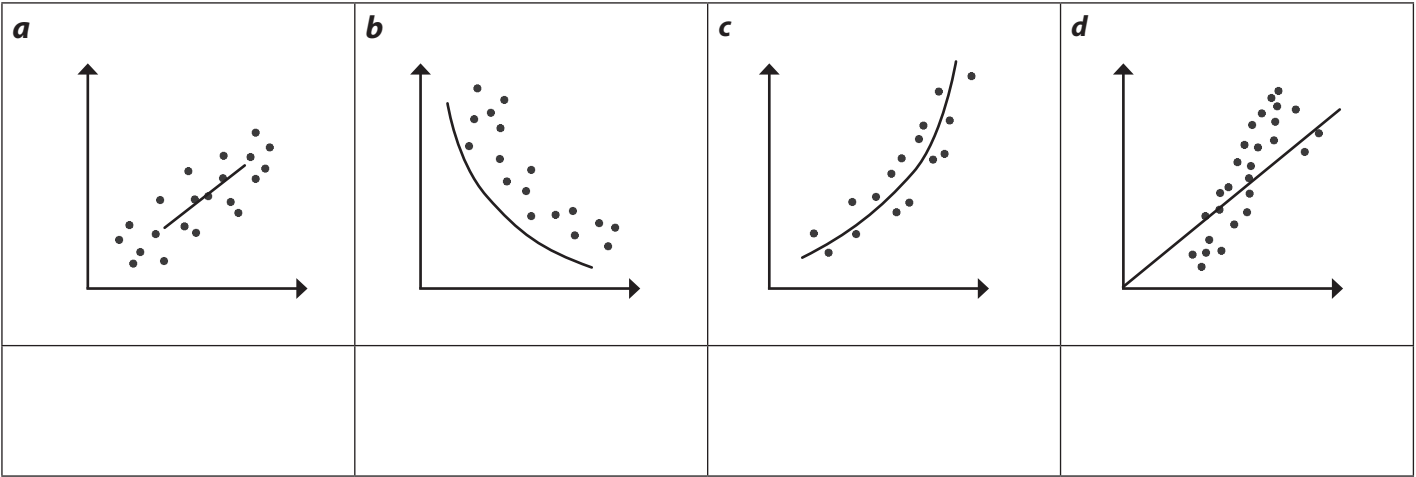
- equal
- first
- hand
- ruler
- straight
- exactly
- last
- not

A line of best fit:

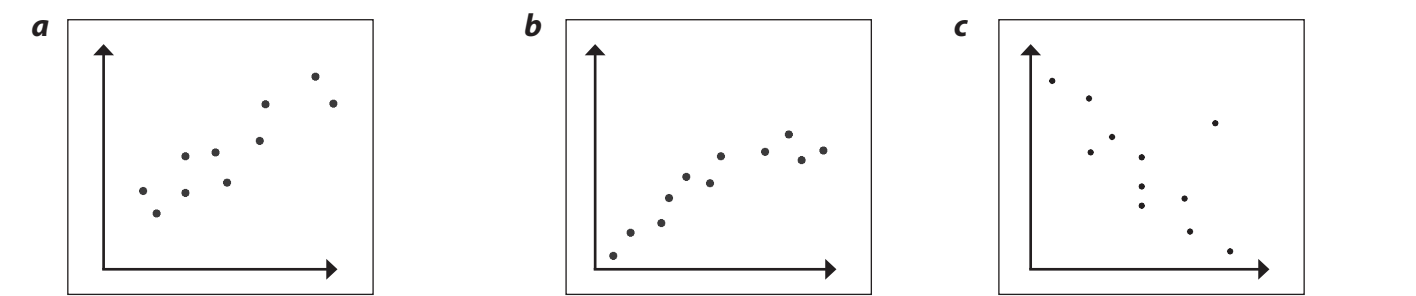
- can be _____, in which case it should be drawn using a _____
- can be curved, in which case it should be drawn smoothly by _____
- should extend just past the _____ and _____ points
- does _____ have to pass through the point (0, 0)
- should have an _____ number of points on either side of the line, if some points do not fall _____ on the line.

2 Tick the scatter graphs which have a correctly drawn line of best fit. Cross the graph if it is not drawn correctly, and in the space below the graph describe how it is incorrect.

In any experiment, there is one set of data where the values are chosen – this is the *independent variable*. The corresponding set of data that is measured is the *dependent variable*.



3 Draw the line of best fit for each scatter graph.



Outliers

In graph c of question 3, one point does not appear to fit the relationship shown by the other points. This point is called an **outlier** and is considered an anomaly, or unusual result. It is usually ignored when drawing the line of best fit because it is probably an inaccurate result.

4 Redraw the line of best fit for the graph in question 3c, if necessary, so that the outlier is excluded.

5 Propose one way an outlier might occur while data is being collected.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Always read all parts of a graph question before drawing the graph, so you can make sure you leave enough space on the grid to extrapolate. Never extrapolate off the grid.

Extrapolating

Extrapolate means to extend a line of best fit **beyond** the plotted points. This allows a prediction to be made for values that lie outside the range of the data.

- 1 Use the steps below to predict the time it takes to read a 350 page book. Show all your working on the graph.

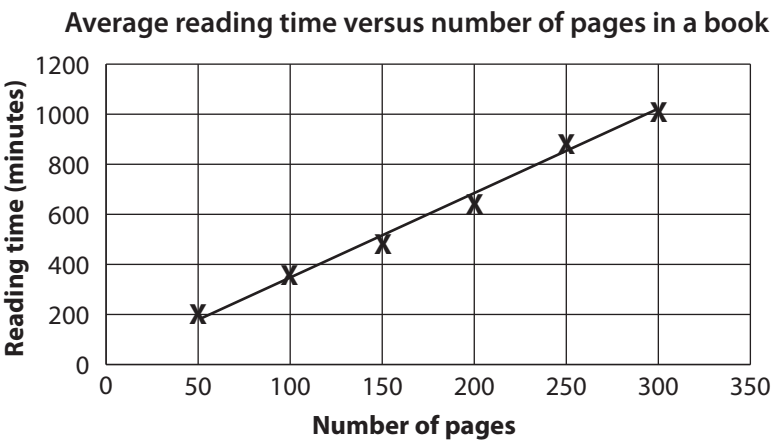
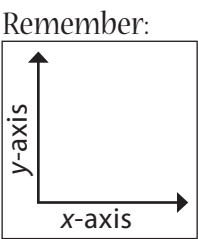
Step 1 Line up your ruler along the line of best fit and extend it to the edge of the grid.

Step 2 Use your ruler to draw a straight line from 350 pages on the x-axis up to the line of best fit. Then draw a line across to the y-axis from where it meets the line of best fit.

Step 3 The place on the scale where the line crosses the y-axis is the answer:

- 2 Predict the number of pages read after 100 minutes. Show all your working on the graph. _____

- 3 What assumption has to be made when a line of best fit is extrapolated? _____



Interpolating

Interpolate means to determine a value from **between** plotted points on a graph. Since the value lies within the range of points, there is no need to extend the line.

- 4 Determine the temperature of the water after 4.5 hours. Show all your working on the graph.

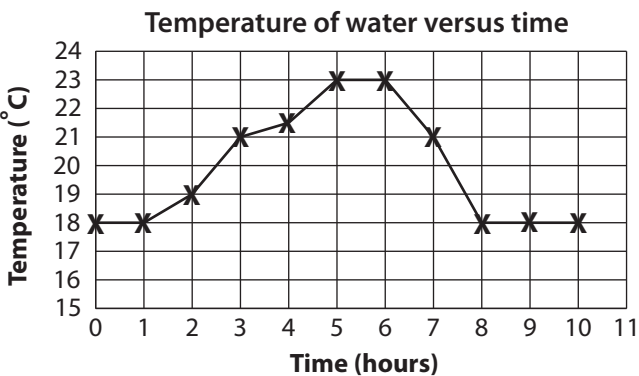
- 5 At what time did the temperature reach 20°C? Show all your working on the graph.

- 6 Explain why only line graphs and scatter graphs can be extrapolated or interpolated.

- 7 Suggest why interpolating is more likely to be accurate than extrapolating.

- 8 Use the graph to predict the water temperature after 11 hours. _____

- 9 Is question 8 an example of interpolating? Explain your answer.

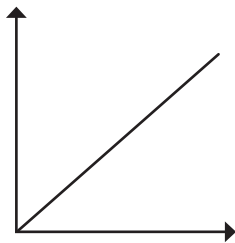


Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

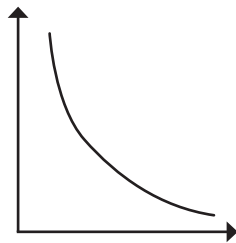
Scatter graphs are drawn to find the relationship between two variables. The shape of a scatter graph's line of best fit is used to identify the relationship and formulate an equation. Ohm's Law is an example of an equation found from the line of best fit of a graph of the current through a resistor and the voltage across it.

m is a constant, meaning its value is always the same. For a straight line, it is the gradient.

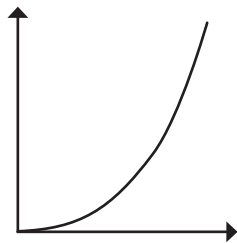
Below are some examples of equations that represent relationships shown by lines of best fit.



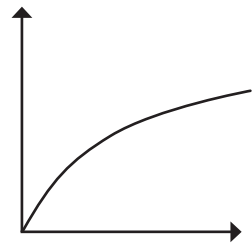
$y = mx$ (linear)



$y = m/x$ (hyperbola)



$y = mx^2$ (parabola)



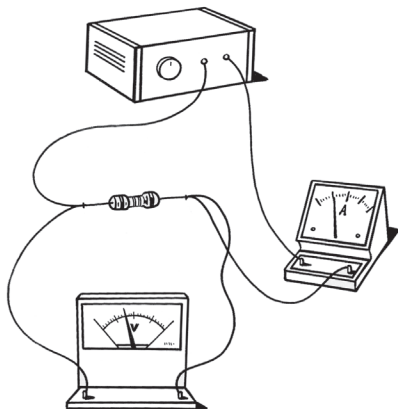
$y = m\sqrt{x}$ (square root)

Determining Ohm's Law

Step 1 Collect the data.

Data is collected on the voltage across a resistor as the current through it is changed. The circuit used to collect the data is shown below.

1 In the box below right, draw a circuit diagram of the equipment shown, using the symbols in the list.



SYMBOLS LIST

ammeter

voltmeter

DC power supply

10Ω (2A rated) resistor

If you have collected your own data by doing this experiment in class, write the data in the table below. Otherwise, use the sample data.

Sample data:

CURRENT, I (A)	VOLTAGE, V (V)
0.20	1.6
0.35	3.4
0.50	5.0
0.67	6.7
0.85	8.6
1.00	10.4

Your data:

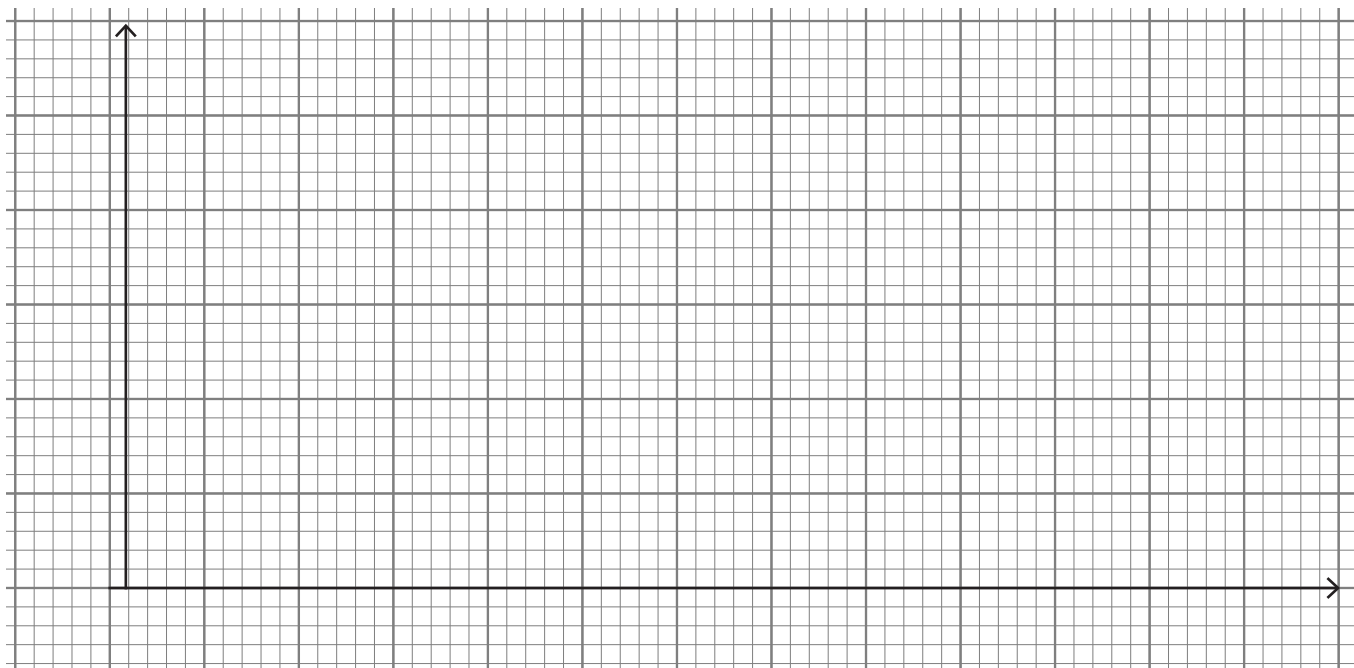
CURRENT, I (A)	VOLTAGE, V (V)

Remember:

2 Identify what (A) and (V) in the table stand for.

Step 2 Graph the data.

- 3** Draw a scatter graph of the data on the grid below. Plot current on the x-axis and voltage on the y-axis.
Hint: Scatter graphs have a line of best fit.

**Step 3** Find the equation.

- 4** Write the equation that represents the relationship shown by the line of best fit. Choose the equation from the examples at the top of the previous page. _____
- 5** On the graph, $y = V$ and $x = I$. In this experiment, the constant m is equal to the resistance of the conductor, R . Substitute these values into the equation you wrote in question 4 and write it in the box below.

where V = voltage measured in _____
 I = _____ measured in _____
 R = _____ measured in ohms, Ω

This formula is known as Ohm's Law.

- 6** The value of the resistance, R , of the conductor can be determined from the gradient of the line of best fit. A gradient is calculated by dividing the rise of the line by the run. The gradient should always be determined from the line of best fit, never from the data in the table.

a Draw a triangle on your line of best fit, similar to the one shown at the right. The triangle can be any size, but the larger the better.

b Write the co-ordinates of the top point (x_2, y_2) . _____

c Write the co-ordinates of the bottom point (x_1, y_1) . _____

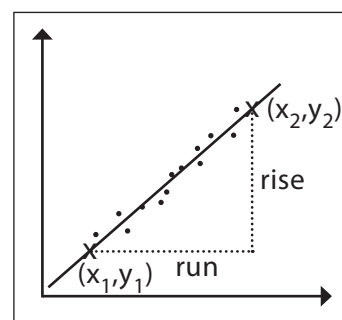
d Resistance, R , = Gradient = $\frac{\text{rise}}{\text{run}} = \frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{\quad}{\quad} = \quad \Omega$

- 7** Does the resistance calculated from the gradient agree with the known value of the resistor? _____

- 8** Calculate the missing values using Ohm's Law and the value for R you determined in question 6d. Show all your working.

a $V = ?$, $I = 1.0\text{A}$ _____

b $V = 15.2\text{V}$, $I = ?$ _____



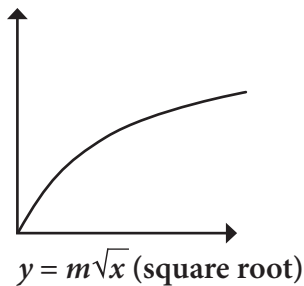
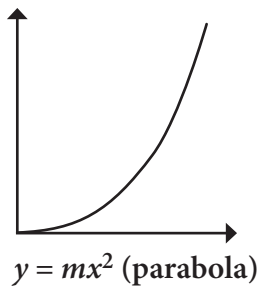
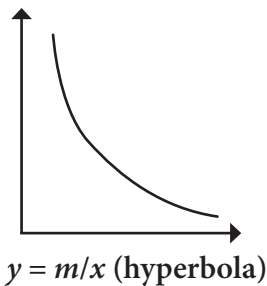
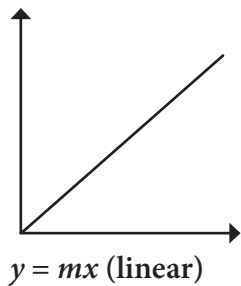
Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Many laws of science have been discovered from the relationships shown by the line of best fit of a scatter graph. One example is the Wave Equation, which describes the relationship between the frequency, wavelength and velocity of waves such as sound and light waves.

m is a constant, meaning its value is always the same. For a straight line, it is the gradient.

Below are some examples of equations that represent relationships shown by lines of best fit.



Determining the Wave Equation

Step 1 Collect the data.

One method of collecting data on the frequency and wavelength of sound waves is to use tuning forks on resonance boxes to produce different waves. The wavelength is four times the length of the resonance box and the frequency is written on the tuning fork.

If you have collected your own data by doing this experiment in class, write it in the table below. Otherwise, use the sample data in the table at the right.

Sample data:

FREQUENCY (Hz)	WAVELENGTH (m)	1/WAVELENGTH (m ⁻¹)
150	2.2	$\frac{1}{2.2} = 0.455$
256	1.29	
384	0.86	
512	0.64	
750	0.44	
1000	0.33	

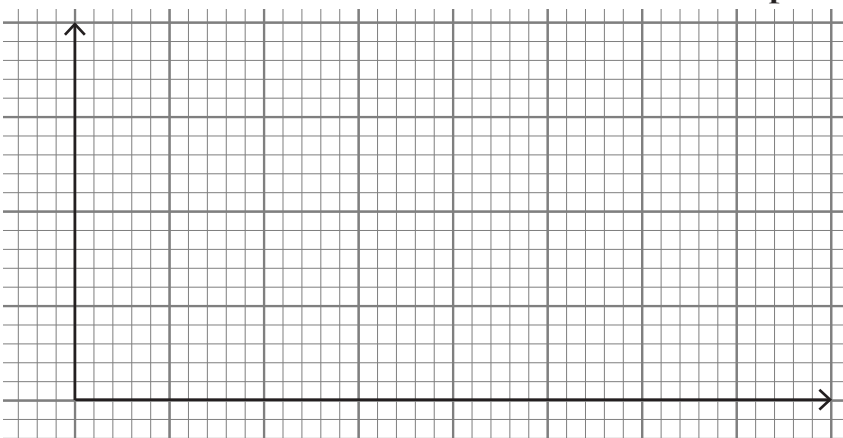
Your data:

FREQUENCY (Hz)	LENGTH OF RESONANCE BOX (m)	WAVELENGTH (m)	1/WAVELENGTH (m ⁻¹)
150	0.55	$4 \times 0.55 = 2.2$	$\frac{1}{2.2} = 0.455$
1000	0.083	0.33	

- 1 Complete the 1/wavelength (m⁻¹) column of the table in Step 1 by calculating the inverse of each wavelength. The first one is done for you.

Step 2 Draw the graph.

- 2 Draw a scatter graph of the data on the grid. Use the sample data if you have not collected your own. Plot frequency on the x-axis and wavelength on the y-axis.



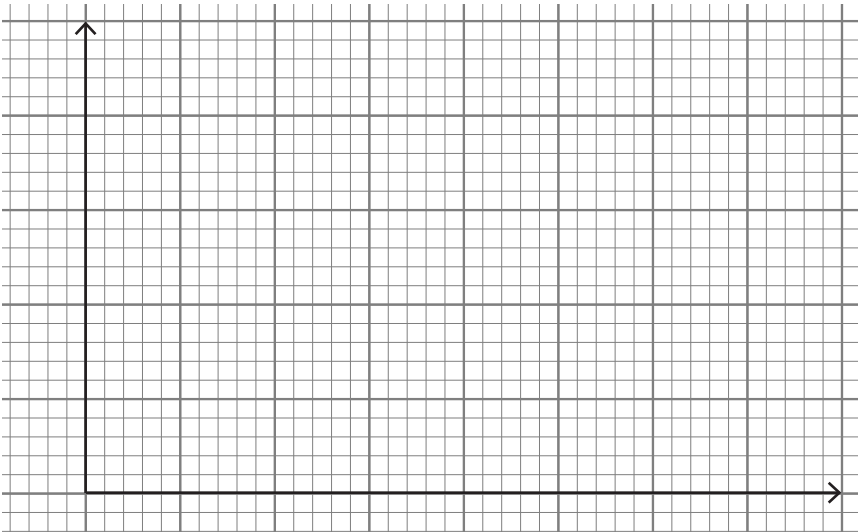
Graph 1

- 3 Write the equation that best describes the relationship between frequency and wavelength shown by the line of best fit. Choose the equation from the examples at the top of the previous page.

The relationship you identified in question 3 can be confirmed by plotting a second scatter graph. This time, plot Inverse Wavelength, also known as $1/\text{wavelength}$, versus Frequency. A straight line of best fit on the graph confirms the relationship identified in question 3.

Graph 2

- 4 Plot a scatter graph of Inverse Wavelength versus Frequency on graph 2. Plot frequency on the x-axis.
- 5 Does the second graph confirm the relationship identified in question 3? *Hint:* Is the line of best fit a straight line?



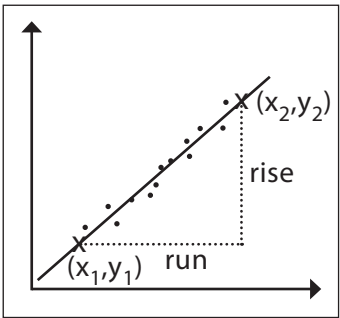
Step 3 Find the equation.

- 6 Write the equation you identified in question 3 that represents the relationship.
- 7 On graph 1, $y = \lambda$ and $x = f$. In this experiment, the constant $m = v$, which is the speed of sound in air. Substitute these values into the equation from question 6 and write it in the box below.

where $v =$ _____ measured in ms^{-1}
 $f =$ _____ measured in _____
 $\lambda =$ _____ measured in _____

This formula is known as the Wave Equation.

- 8 Calculate the speed of the sound waves from the gradient of graph 2. The gradient should always be determined from the line of best fit, never from the data in the table.
- a Draw a triangle on your line of best fit, similar to the one shown at the right. The triangle can be any size, but the larger the better.
- b Write the co-ordinates of the top point (x_2, y_2) . _____
- c Write the co-ordinates of the bottom point (x_1, y_1) . _____
- d Gradient = $\frac{\text{rise}}{\text{run}} = \frac{(y_2 - y_1)}{(x_2 - x_1)} =$ _____ $=$ _____ ms^{-1}
- e Because the gradient is determined from the inverse graph, the answer you found in part d needs to be inverted:
- Speed, $v = \frac{1}{\text{gradient}} =$ _____ ms^{-1}



- 9 Use the Wave Equation and the speed you determined in question 8e to calculate the unknown values in the table.

FREQUENCY (Hz)	WAVELENGTH (m)
a 1 230	
b	0.036
c 96.1×10^6	

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

You need a protractor to complete this activity.

1 For Nivi’s school project, she needed to find out which metals are present in the alloys used to make ‘silver’ and ‘gold’ coins. She found pie graphs of the information, but they weren’t labelled with the amounts.

- a Which type of coin contains a higher percentage of copper? _____
- b Nivi weighed 15 ‘silver’ 20c coins and found the average mass was 6.81 g. Determine the mass of each metal in the coin and write it in the table.

Hint: Measure the angle for each metal, then work out the mass using:

$$\frac{\text{angle}}{360^\circ} \times \text{total mass of coin}$$

2 Use the information in the graph ‘Sources of electricity generation in Australia in 1999’ to answer the following questions.

- a What percentage of electricity generation came from:
- ▲ renewable sources? _____
 - ▲ coal? _____
- b Which energy source provided:
- ▲ the smallest proportion of Australia’s electricity generation?

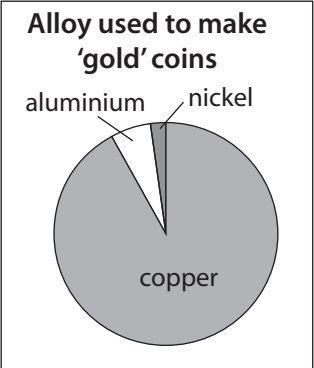
▲ the largest proportion of Australia’s electricity generation? _____

3 The graph ‘Amount of C, H and O by mass’ shows the amount of carbon, hydrogen and oxygen present in three common compounds.

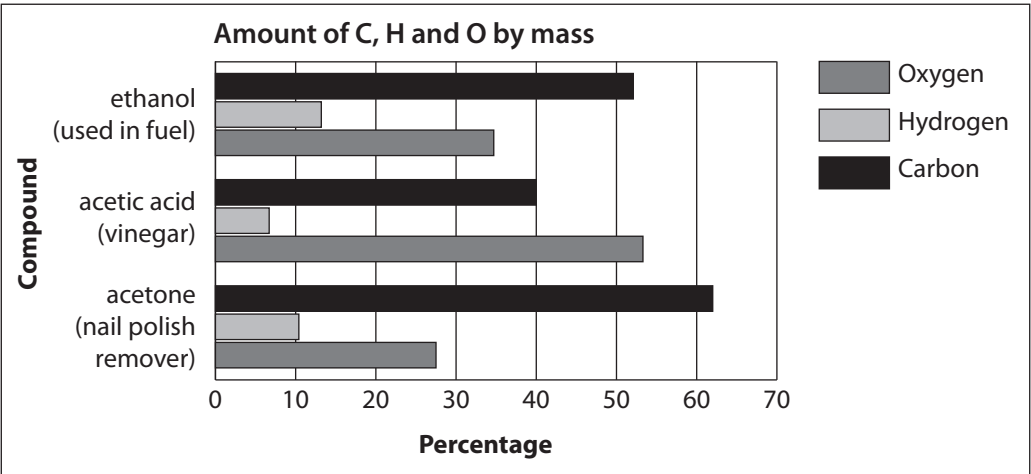
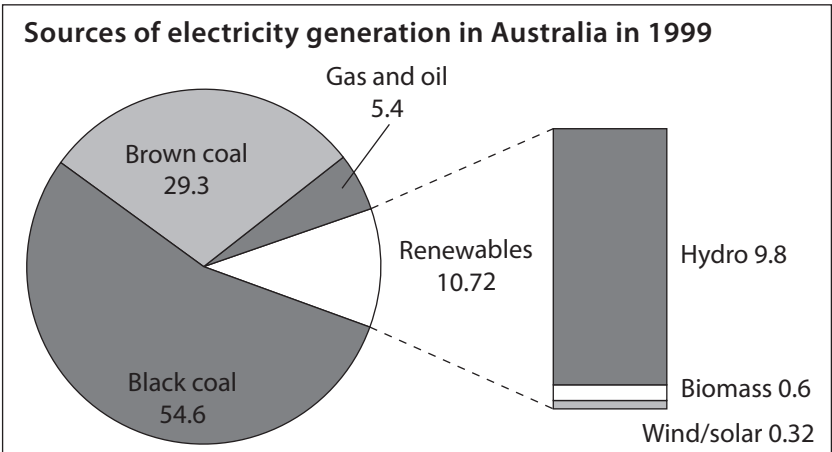
- a Identify the compound with the least amount of carbon.

b Propose a reason for leaving spaces between the bars for different compounds, but no spaces between the bars for the data for the same compound.

- c Which element makes the least contribution to the mass of these compounds?

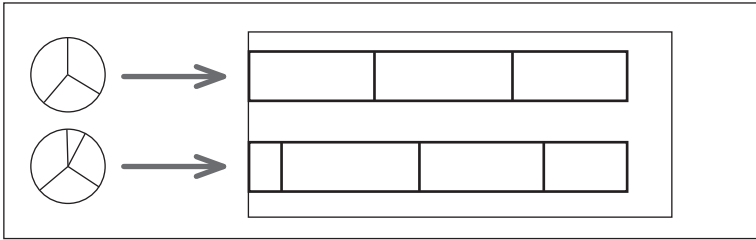


METAL	ANGLE (°)	MASS (g)
Copper		
Nickel		

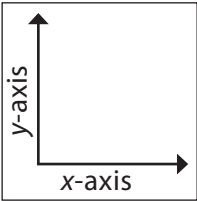


Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

A divided bar graph is several pie graphs stretched into the bars of a bar graph.



Remember:



A pie graph is divided into sectors, but each bar on a divided bar graph is divided into rectangles. The lengths of the rectangles show the size of each proportion.

1 Follow the steps to draw a divided bar graph of Omar and Patrick’s ‘Average time spent on activities each weekday’.

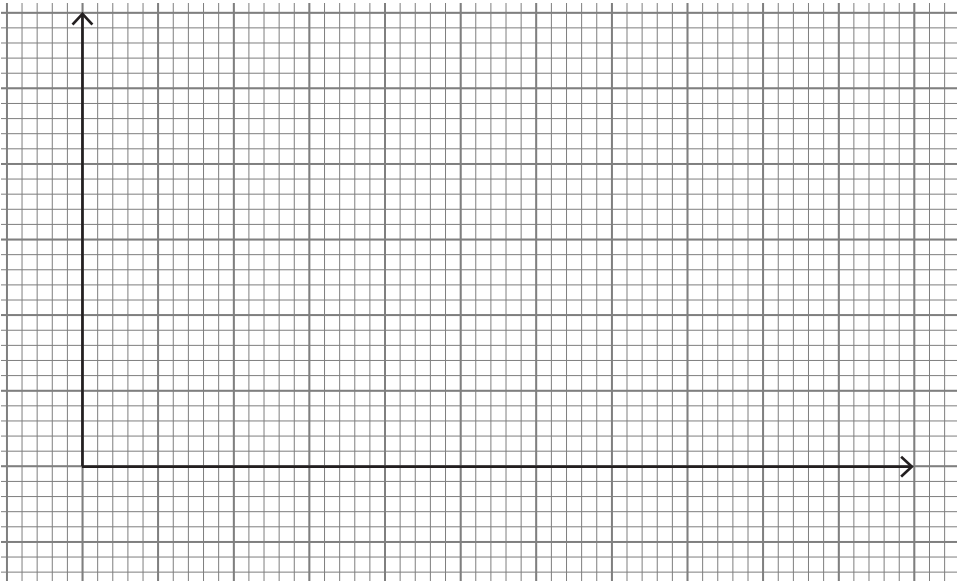
Step 1 Find a suitable scale for the x-axis on the grid provided below. If you use 1 cm = 10%, the x-axis would be 10 cm long – a good length for the available grid space.

Step 2 Use the scale to calculate the rectangle length for each of Omar and Patrick’s activities using the scale. For example, the percentage for ‘Omar – Sleeping’ is: 8 hours ÷ 24 hours × 100 = 33%. Each percentage can be rounded off to a whole number. Since 1 cm = 10%, the rectangle length will be 33 ÷ 10 = 3.3 cm.

2 Complete the table for the rest of Omar and Patrick’s activities.

	OMAR			PATRICK		
Activity	Time (hours)	Percentage	Length (cm)	Time (hours)	Percentage	Length (cm)
Sleeping	8	33	3.3	8.5		
School	7			7		
Homework	1			2		
Watching TV/ Internet	5			3		
Other	3			3.5		
TOTAL	24					

Step 3 Starting at 0, mark the scale 1 cm = 10% along the x-axis. Label the x-axis with ‘Percentage’. Label the y-axis with the students’ names, leaving a gap between each name and the x-axis. Write a title for the graph.



KEY

☐

☐

Step 4 Draw a 10 cm long bar starting from zero for Omar. Measure 3.3 cm along the bar from zero and draw a line across it. This is the ‘Sleeping’ rectangle. Measure and draw the rest of Omar’s rectangles by dividing up the bar. The bar should look similar to the ones at the top of the previous page. Repeat this step to create Patrick’s bar.

Step 5 Colour Omar and Patrick’s ‘Sleeping’ rectangles the same colour. Colour the first box in the Key with this colour and label it ‘Sleeping’. Complete the key using different colours for each activity.

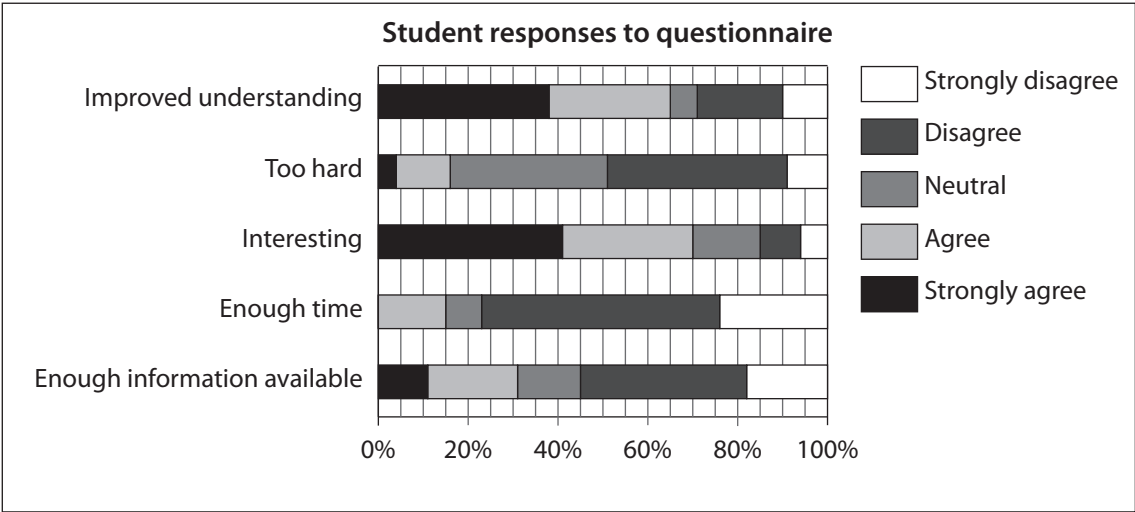
3 Swap graphs with the person next to you. Mark their graph using the criteria in the table.

CRITERIA	MARK
x-axis scale increases by equal amounts.	
x-axis scale starts at 0.	
Graph title describes the data in the table.	
x-axis title is ‘Percentage’.	
y-axis title is ‘Student’ with Omar and Patrick’s names.	
Rectangle lengths are drawn correctly.	
All activities are listed in the key.	
Each activity is the same colour and different activities have different colours.	
Equal spaces between the bars and x-axis.	
Graph takes up maximum grid space.	
TOTAL	/10

A good *title* for a divided bar graph is a description of what the graph is showing.

- 4** Use your graph to answer the following questions.
- a** Which student spends more time on homework? _____
- b** How much more time does Omar spend watching TV and on the Internet than Patrick? _____
- c** What activities could be included in the ‘Other’ category?

5 After his class had finished a homework assignment, a teacher gave them a survey on their experiences while completing it. The graphed results are shown below.



- a** What did students think about how hard the assignment was?

- b** What percentage of students thought the assignment was interesting? _____
- c** Identify one way the teacher could improve his graph.

- d** Describe how the teacher could improve the assignment. _____



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

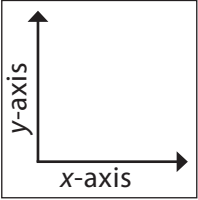
- Composite line graphs are drawn to show:
- two or more sets of data on the same variable
 - or
 - data on two or more different variables.

1 Draw a composite line graph of the height of three siblings from ages 10–18 years, using the data shown in the table at the right.

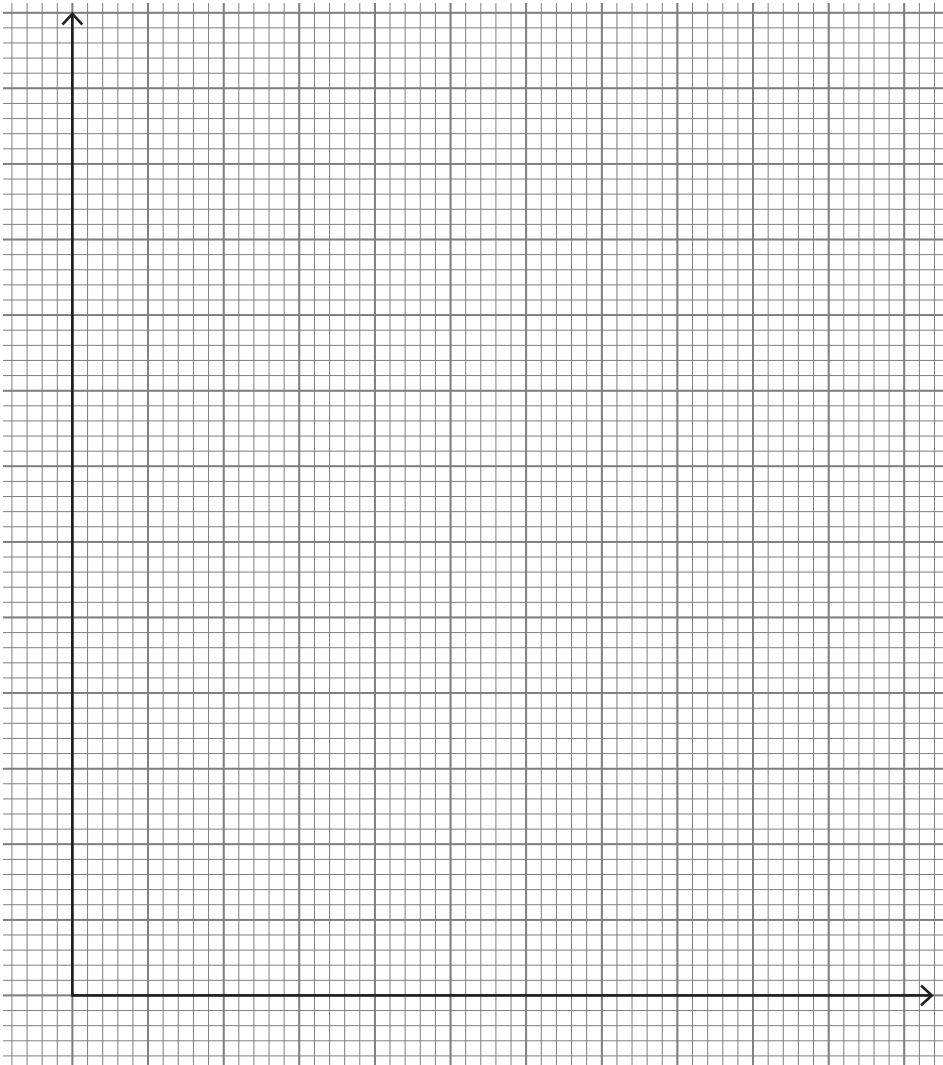
Step 1 Construct one graph with ‘Age’ on the x-axis and ‘Height’ on the y-axis.

Step 2 Draw a line graph using the data for Sarah in one colour. Colour in the box for Sarah in the key with the same colour. Repeat this step to draw two more lines on the same graph, one for each of the other two children. Use a different colour for each line. Add their colours and names to the key.

Remember:



Age (years)	HEIGHT (m)		
	Sarah	Simon	Mark
10	1.55	1.42	1.41
11	1.62	1.49	1.46
12	1.69	1.57	1.51
13	1.75	1.64	1.59
14	1.77	1.72	1.67
15	1.77	1.77	1.82
16	1.77	1.81	1.87
17	1.77	1.81	1.92
18	1.77	1.81	1.92



KEY

☐ Sarah

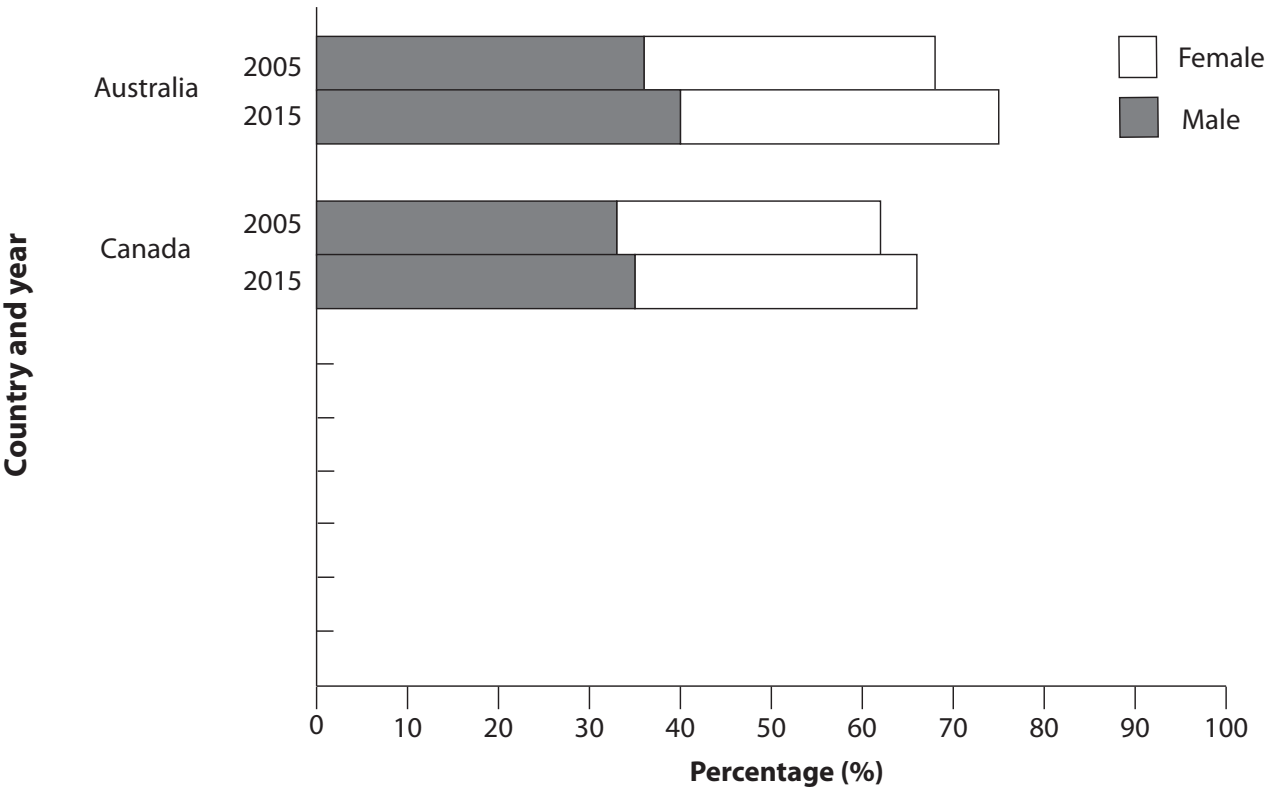
- 2 a Which child had the most rapid growth spurt? At what age? _____
- b Justify your answer to part a. _____
- _____

3 Complete the table at the right using the information in the composite bar graph. The graph shows the percentage of overweight or obese people aged 15 years and older from four countries in 2005, and the predicted percentage in 2015.



COUNTRY	YEAR	PERCENTAGE OF THE POPULATION	
		Male	Female
Australia	2005		
	2015		
Canada	2005		
	2015		
UK	2005	33	31
	2015	35	33
USA	2005	38	37
	2015	43	40

Percentage of overweight or obese people aged 15 years and older in 2005 and 2015

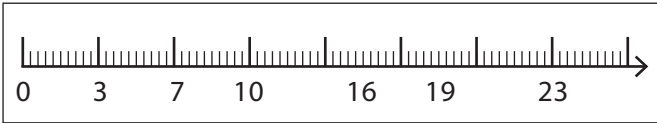


- 4 Complete the composite bar graph using the information in the table.
- 5 In 2015, which group of people will be most at risk from weight-related diseases?
- 6 Treating weight-related diseases costs society a lot of money. Suggest one way governments could act to reduce the obesity epidemic.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

1 Identify one error made when this graph scale was drawn.



2 Suzanne placed increasing amounts of mass on a spring and measured how far it stretched. She plotted a scatter graph of her results. Write an appropriate title for her graph.

3 Compare column graphs and line graphs. *Hint: 'Compare' means describe similarities and differences.*

4 Justify the type of graph you would use to represent the data in the table at the right.

Hint: 'Justify' means identify which graph you would choose then state the reason(s) for choosing it.

LOCATION OF WATER IN EARTH'S HYDROSPHERE	PERCENTAGE (%)
Ocean	97
Ice caps/glaciers	2.997
Fresh water	0.003

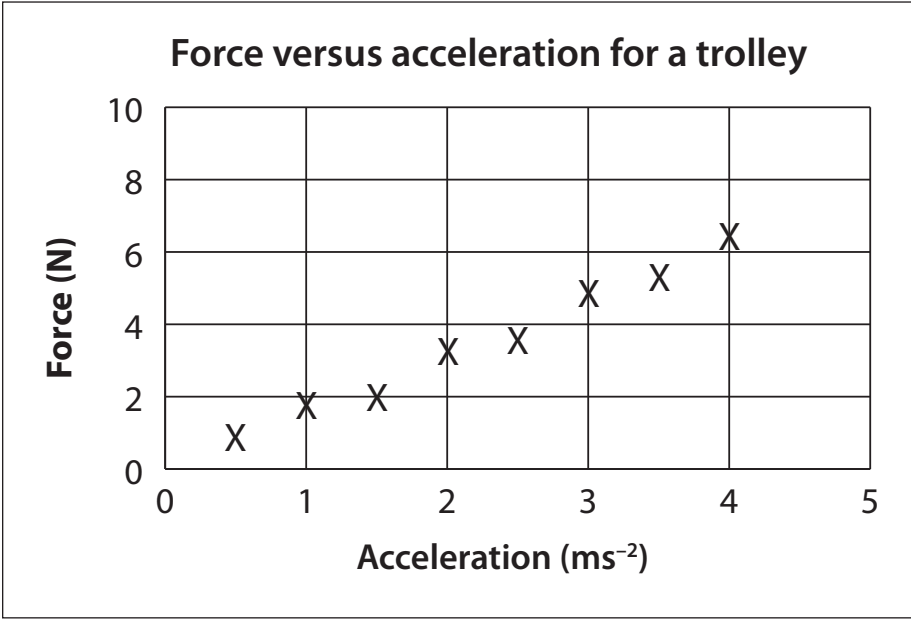
5 The following questions refer to the 'Force versus Acceleration for a trolley' graph.

a Draw the line of best fit on the graph.

b Determine the acceleration if a force of 7N is applied. Show your working on the graph.

c Identify the term that describes the technique you used in part b.

d Determine the force if the acceleration is 2.25 ms⁻². Show your working on the graph.



e Identify the term that describes the technique you used in part d. _____

f Determine the mass in kilograms of the object by calculating the gradient of the graph.

g Write the equation for the relationship between force, mass and acceleration shown on the graph.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Microsoft Excel is a program used to organise data, perform calculations and create graphs.

Excel is made up of worksheets that look like large tables. Each column has a unique letter, and each row has a unique number. Each rectangle is called a cell and each cell has unique co-ordinates based on its column and row.

1 On the sample Excel worksheet shown:

- a circle column D.
- b identify what is typed in cell A3. _____
- c draw a triangle in cell B6.
- d draw a line through row 5.

2 Follow the steps below to create a pie graph of 'The percentage of different gases in Earth's atmosphere' using the data shown in the sample Excel worksheet at the right.

Hints:

- Never leave cells within the data table blank. If text is too long for a cell, move the mouse over the border between the column labels A and B so this symbol appears:  Click and hold the left mouse button and drag the column wider. Rows can also be made taller this way.
- To edit the data typed into a cell, first double-click on the cell. With only one click, everything in the cell will be overwritten.

APPLE MAC ALERT!

In the steps below, replace:

- 'right-click' with 'hold down the Ctrl key and click'
- 'hold down the Ctrl key' with 'hold down the apple/command key'.

Sample Excel worksheet

	H15				
	A	B	C	D	E
1	Gas	Percentage			
2	Nitrogen	78			
3	Oxygen	21			
4	Argon	0.9			
5					
6					
7					
8					

Create a data table

Step 1 Enter the data in the sample Excel worksheet into your own Excel worksheet, exactly as shown.

Step 2 Check that the data in column B adds up to 100%, otherwise Excel will change the values to force them to add up to 100%. Do this by:

Excel 2003/Excel Mac 2004 Clicking on cell B5, then clicking on 'Insert', then 'Function'. In the 'Most Recently Used' category, click 'SUM'. If SUM is not listed, find it in the 'All' category. Click 'OK'. A window will pop up with 'Number 1 B2:B4'. Click 'OK'.	Excel 2007/Excel Mac 2008 Clicking on cell B5, then clicking on 'Σ AutoSum'. Flashing lines should appear automatically around the numbers you want to add up. Check the range B2:B4 is selected, then click 'Enter'.
---	--

Step 3 When the data does not add up to 100%, the missing amount is usually called 'Other'. Add this 'Other' amount to the table by:

Excel 2003/Excel Mac 2004 Inserting a new row. Click on the label for row 5, then click on 'Insert', then 'Rows'.	Excel 2007/Excel Mac 2008 Inserting a new row. Click on the label for row 5, then in the 'Home' tab, 'Cells' section, click on 'Insert'.
--	---

Type 'Other' into cell A5 and the missing amount into cell B5. Press Enter and the Sum should now equal 100. If it does not, check the 'Other' amount has been calculated correctly.

Step 4 Double-click on the 'Sheet1' tab at the bottom of the screen and rename it 'Pie graph'. Save the Excel file by clicking on the disk icon. Name the file 'Pie graph', then choose a folder to save it in and click 'OK'.

Create the graph

Step 5 Highlight cells A2 to B5 by clicking on A2 and, holding down the mouse button, dragging to cell B5.

Excel 2003/Excel Mac 2004 Step 6 Click on the Chart Wizard and choose 'Pie chart'. Choose 'chart sub-type 1'. Click 'Next'. Step 7 A preview of your graph will appear. If it looks right, click 'Next'. If not, there may be something wrong with the way the data is entered or selected, so click 'Cancel' and fix or re-select the data. Step 8 • In the 'Title' tab, type the chart title. • As labels will be added to the graph, it does not need a legend. Uncheck 'Show Legend' in the 'Legend' tab. • In the 'Data Labels' tab, check 'Category name' and 'Percentage'. Click 'Next'. Click 'Finish'. Step 9 • Click on the labels and drag each one outwards so it is readable. Use the Undo button if needed! • Make the graph bigger by clicking on the plot area or the middle of the graph frame and dragging it taller. • If necessary, adjust the number of decimal places displayed by right-clicking on one data label and choosing 'Format Data Labels...'. In the 'Number' tab, adjust the number of decimal places.	Excel 2007/Excel Mac 2008 Step 6 In the 'Insert' tab, click on 'Pie' chart. Choose 'chart 2D-Pie sub-type 1'. Step 7 In the 'Chart Tools – Design' tab, in the 'Chart Layouts' section, choose chart sub-type 1. Step 8 Double-click on the 'Chart Title' and type in a title. Step 9 • If necessary, adjust the number of decimal places displayed by clicking on any of the percentages, then right-clicking and choosing 'Format Data Labels'. In the 'Number' section, choose 'Percentage' and adjust the decimal places. Click 'Close'. • Adjust the height of the graph to allow more room for the labels by clicking on the top middle of the graph frame and dragging it taller. • If more room is needed to display the data labels, click on each one twice and drag it into position. Use the Undo button if needed! A good title for a pie graph is a description of what the graph is about. Don't forget to save your work regularly.
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Customise and print

Step 10 Add your name so you know which graph is yours when you print it. Do this by:

Excel 2003/Excel Mac 2004 • Clicking on the 'Tools' menu and selecting 'Customise...'. In the 'Toolbars' tab, check the 'Drawing' box and click 'Close'. In the Drawing menu that has just appeared at the bottom of the page, click on the text box icon, then click on an empty space on your graph. Type in your name.	Excel 2007/Excel Mac 2008 • Clicking on a blank corner of the graph window. In the 'Chart Tools', 'Layout' tab click on 'Text box', then click on an empty space on the graph window. Type in your name.
---	--

Step 11 Change the appearance of your graph. First select the graph by clicking on it, then right-click on different parts of the graph. Options are available to customise the background colours, the fonts used and more.

Step 12 Print Preview can save a lot of printing money and the environment by not wasting paper. It shows what will be printed before it is sent to the printer. Print the graph by:

Excel 2003/Excel Mac 2004 • Clicking on the graph, then click on Print Preview. If your graph looks correct and takes up the whole page, click 'Print'. Otherwise click 'Close' and change it.	Excel 2007/Excel Mac 2008 • Clicking on the graph, then click on 'Print Preview'. 'Print Preview' is located in the top left Office button under 'Print'. If your graph looks correct and takes up the whole page, click 'Print'. Otherwise, click 'Close' and change it.
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Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Step 13 Save and close your file. Paste the graph into your book.

Microsoft Excel is a program used to organise data, perform calculations and create graphs.

Excel is made up of worksheets that look like large tables. Each column has a unique letter, and each row has a unique number. Each rectangle is called a cell and each cell has unique co-ordinates based on its column and row.

- 1 Follow the steps to create a bar graph on the 'Nutritional information of peanut butter', using the data in the sample Excel worksheet.

Bar graphs and divided bar graphs are the only types of graphs to have the first column in a data table plotted on the *y*-axis.

Create a data table

- Step 1 Enter the data in the sample Excel worksheet into your own Excel worksheet, exactly as shown.
- Step 2 Check the data adds up to 25 g. Do this by:

APPLE MAC ALERT!

In the steps below, replace:

- 'right-click' with 'hold down the Ctrl key and click'
- 'hold down the Ctrl key' with 'hold down the apple/command key'.

Sample Excel worksheet

	M15		
	A	B	C
1	Nutrient	Grams per 25g	
2	Protein	6.1	
3	Fat	13.5	
4	Carbohydrates	2.4	
5	Fibre	1.6	
6			
7			

Excel 2003/Excel Mac 2004 • Clicking on cell B6, then clicking on 'Insert', then 'Function'. In the 'Most Recently Used' category, click on 'SUM'. If SUM is not listed, find it in the 'All' category. Click 'OK'. A window will pop up with 'Number 1 B2:B5'. Click 'OK'.	Excel 2007/Excel Mac 2008 • Clicking cell B6, then clicking on 'Σ AutoSum'. Flashing lines should appear automatically around the numbers you want to add up. Check the correct range B2:B5 is selected. Click 'Enter'.
---	---

- Step 3 When the data does not add up, the missing amount is usually called 'Other'. Add the 'Other' amount to the table by:

Excel 2003/Excel Mac 2004 • Inserting a new row. Click on the label for row 6, then click on 'Insert' then 'Rows'.	Excel 2007/Excel Mac 2008 • Inserting a new row. Click on the label for row 6, then in the 'Home' tab, 'Cells' section, click on 'Insert'.
--	--

Type 'Other' into cell A6 and the missing amount into cell B6. Press Enter and the Sum should now equal 25. If it does not, check the 'Other' amount has been calculated correctly.

Formulas in Excel + means 'add' - means 'subtract' * means 'multiply' / means 'divide' = means 'equals'	\$B\$5 is an absolute reference. It means 'if a formula is copied, always use the data from this cell'. In the case of the sample data, it would mean always using 1.6.	B2, without the \$ signs, is a relative reference. It means 'if a formula is copied, use the data from the cell which is in the same position relative to this cell'. In the case of the sample data, it means use B2 for protein, B3 for fat, and so on.
--	--	--

- 2 a Write the absolute reference for the cell containing the AutoSum. _____
- b Write the relative reference for the cell containing the word 'Protein'. _____

- Step 4 Excel can be used to calculate percentages and other values from data, much more quickly than using a calculator. Type the title 'Percentage' into cell C1. Enter the formula =(B2/\$B\$7)*100 into cell C2 and press Enter. Rather than entering this formula every time, Excel can copy it for all the other data. Click on cell C2 and move the mouse over the bottom right corner of the cell until it forms a small black cross. Click on the cross and drag it downwards to cell C6. Let go of the mouse button.

Create the graph

A good title for a bar graph is a description of what data is being graphed.

Step 5 Highlight cells A2 to A6 and C2 to C6. To highlight columns that are not next to each other, highlight cells A2 to A6, let go of the mouse button, hold down the Ctrl key, then highlight cells C2 to C6. Let go of the mouse button and the Ctrl key.

Excel 2003/Excel Mac 2004 Step 6 Click on the Chart Wizard and choose 'Bar sub-type 1'. Click 'Next', and click 'Next' again. Step 7 • In the 'Title' tab, add a chart title and titles for the (X) and (Y) axes. Don't forget to include any units. • Change the gridlines in the 'Gridlines' tab to what looks best. • Simple bar graphs do not need a legend. Remove the legend in the 'Legend' tab. • Click 'Next'. Click 'Finish'.	Excel 2007/Excel Mac 2008 Step 6 Click on the 'Insert' tab and click on 'Bar'. Choose '2D-Bar sub-type 1'. Step 7 • In the 'Chart Tools', 'Layout' tab, click on 'Axis Titles' and add a Primary Horizontal Axis Title. Also add a rotated Primary Vertical Axis Title. Double-click on each and type appropriate titles. Don't forget to include any units. • Click on 'Chart Title' and choose 'Above Chart'. Double-click on the newly created chart title and type in a title. • Simple bar graphs do not need a legend. Click on 'Legend' and choose 'None'.
---	---

Step 8 Double-click on the 'Sheet1' tab at the bottom of the screen and rename it 'Bar graph'. Save the Excel file by clicking on the disk icon. Name the file 'Bar graph', then choose a folder to save it in. Click 'OK'.

Customise and print

Step 9 Add your name so you know which graph is yours when you print it. Do this by:

Excel 2003/Excel Mac 2004 • Clicking on the 'Tools' menu and selecting 'Customise...'. In the 'Toolbars' tab, check the 'Drawing' box and click 'Close'. In the Drawing menu that has just appeared at the bottom of the page, click on the text box icon, then click on an empty space on your graph. Type in your name.	Excel 2007/Excel Mac 2008 • Clicking on a blank corner of the graph window. In the 'Chart Tools', 'Layout' tab click on 'Text box', then click on an empty space on the graph window. Type in your name.
---	--

Step 10 Print Preview can save a lot of printing money and the environment by not wasting paper. It shows what will be printed before it is sent to the printer. Print the graph by:

Excel 2003/Excel Mac 2004 • Clicking on the graph, then click on Print Preview. • If your graph looks correct and takes up the whole page, click 'Print'. Otherwise, click 'Close' and change it.	Excel 2007/Excel Mac 2008 • Clicking on the graph, then click on 'Print Preview'. 'Print Preview' is located in the top left Office button under 'Print'. If your graph looks correct and takes up the whole page, click 'Print'. Otherwise, click 'Close' and change it.
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Step 11 Save and close the file. Paste the graph into your book.

3 a Type the data shown at the right into a worksheet and perform the SUM shown in cell A8.

b What do cells A1 to A7 add up to? _____

4 a If '=A4/\$A\$7' was typed into cell B4, what would the answer be? _____

b If this formula was copied to cell B5, what would the answer be? _____

SUM		X	✓	<i>f_x</i>	=SUM(A1:A7)
	A	B			
1	57				
2	98				
3	64				
4	33				
5	27				
6	101				
7	3				
8	=SUM(A1:A7)				
9					



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Excel 2003/Excel Mac 2004

- Click on the 'Series' tab. Click on 'Series 1' and click 'Remove'. Click on the 'Category (X) axis labels:' text box, then highlight the x-axis data. Click 'Next'.

Step 6

- In the 'Title' tab, type the chart title and the titles for the (X) and (Y) axes. Don't forget to include any units.
- Change your gridlines in the 'Gridlines' tab to what looks best.
- Remove the legend in the 'Legend' tab.
- Click 'Next'. Click 'Finish'.

A good title for a column graph would be a brief sentence describing the data.

Excel 2007/Excel Mac 2008

- Click on the graph, then in the 'Chart Tools – Design' tab, click on 'Select Data'. Under 'Horizontal (Category) Axis Labels' click on 'Edit'. Highlight the column A data and click 'OK'.
- Under 'Legend Entries (Series)', click on 'Series 1' then click on 'Remove'. Click 'OK'.

Step 6

- In the 'Chart Tools', 'Layout' tab, click on 'Axis Titles' and add a Primary Horizontal Axis Title. Also add a rotated Primary Vertical Axis Title. Double-click on each and type appropriate titles. Don't forget to include any units.
- Click on 'Chart Title' and choose 'Above Chart'. Double-click on the newly created chart title and type in a title.
- Click on 'Legend' and choose 'None'.

Customise and print

Step 7 Add your name so you know which graph is yours when you print it. Do this by:

Excel 2003/Excel Mac 2004

- Clicking on the 'Tools' menu and selecting 'Customise...'. In the 'Toolbars' tab, check the 'Drawing' box and click 'Close'. In the Drawing menu that has just appeared at the bottom of the page, click on the text box icon, then click on an empty space on your graph. Type in your name.

Excel 2007/Excel Mac 2008

- Clicking on a blank corner of the graph window. In the 'Chart Tools', 'Layout' tab, click on 'Text box', then click on an empty space on the graph window. Type in your name.

Step 8 Change the appearance of your graph. First select the graph by clicking on it, then right-click on different parts of the graph. Options are available to customise the background colours, the fonts used, and more.

Step 9 Print Preview can save a lot of printing money and the environment by not wasting paper. It shows what will be printed before it is sent to the printer. Print the graph by:

Excel 2003/Excel Mac 2004

- Clicking on the graph, then click on Print Preview.
- If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.

Excel 2007/Excel Mac 2008

- Clicking on the graph, then click on 'Print Preview'. 'Print Preview' is located in the top left Office button under 'Print'. If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.

Step 10 Save and close the file. Paste the graph into your book.

3 a Practise creating superscripts and subscripts by entering the data shown into your own Excel worksheet.

b In your workbook, compare the atmospheres of Earth and Mars. *Hint:* 'Compare' means describe similarities and differences, usually at least two of each. Use words such as 'and', 'however', 'both' and 'whereas'.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



J17		fx		
	A	B	C	D
		Percentage on Mars	Percentage on Earth	
1	Gas			
2	CO ₂	95.3	0.03	
3	N ₂	2.7	78	
4	Ar	1.6	0.9	
5	O ₂	0.13	21	
6	CO	0.07	variable ¹	
7	H ₂ O	0.03	variable ¹	
8	other	0.17	0.07	
9				
10	¹ varies considerably in different locations due			
11	to local weather and pollution			

Microsoft Excel is a program used to organise data, perform calculations and create graphs.

Excel is made up of worksheets that look like large tables. Each column has a unique letter, and each row has a unique number. Each rectangle is called a cell and each cell has unique co-ordinates based on its column and row.

- 1 On the sample Excel worksheet shown:
- a draw a cross in cell C12.
 - b identify what is typed in cell B1. _____
 - c identify why the border of cell B8 looks different.
- _____
- _____

- 2 Follow the steps to create a histogram of ‘Frequency of a population of Green Tree frogs’ using the data shown in the sample Excel worksheet.

Hints:

- Never leave cells within the data table blank. If text is too long for a cell, move the mouse over the border between the column labels A and B so this symbol appears:  Click and hold the left mouse button and drag the column wider. Rows can also be made taller this way.
- To edit the data typed into a cell, first double-click on the cell. With only one click, everything in the cell will be overwritten.

APPLE MAC ALERT!

In the steps below, replace:

- ‘right-click’ with ‘hold down the Ctrl key and click’
- ‘hold down the Ctrl key’ with ‘hold down the apple/command key’.

Sample Excel worksheet

	B8			
	A	B	C	D
1	Month	Frequency		
2	Jan	16		
3	Feb	13		
4	Mar	13		
5	Apr	9		
6	May	6		
7	Jun	6		
8	Jul	7		
9	Aug	7		
10	Sep	26		
11	Oct	26		
12	Nov	19		
13	Dec	16		

Create a data table

- Step 1 Enter the data in the sample Excel worksheet into your own Excel worksheet, exactly as shown.
- Hint: Excel can complete a series if the first few values are entered. Type in ‘Jan, Feb, Mar’ into cells A2, A3 and A4 respectively, then highlight them and move the mouse over the bottom right corner of the selection until it forms a small black cross. Click on the cross and drag it downwards until Dec appears. Let go of the mouse button.

- Step 2 Double-click on the ‘Sheet1’ tab at the bottom of the screen and rename it ‘Histogram’. Save the Excel file by clicking on the disk icon. Name the file ‘Histogram’, then choose a folder to save it in and click ‘OK’.

Create the graph

- Step 3 Highlight cells A2 to B13 by clicking on A2 and, holding down the mouse button, dragging to cell B13.

Don’t forget to save your work regularly.

Excel 2003/Excel Mac 2004	Excel 2007/Excel Mac 2008
Step 4 Click on the Chart Wizard and choose ‘Column chart sub-type 1’. Click ‘Next’. Step 5 If two sets of columns have appeared, follow this step. Otherwise, skip to Step 6. Click on the ‘Series’ tab. Click on ‘Series 1’ and click ‘Remove’. Click on the ‘Category (X) axis labels:’ text box, then highlight the x-axis data. Click ‘Next’.	Step 4 Click on the ‘Insert’ tab, then ‘Column’. Choose ‘2D-column chart sub-type 1’. Step 5 If two sets of columns have appeared, follow this step. Otherwise, skip to Step 6. • Click on the graph, then in the ‘Chart Tools – Design’ tab, click on ‘Select Data’. Under ‘Horizontal (Category) Axis Labels’ click on ‘Edit’. Highlight the column A data and click ‘OK’. • Under ‘Legend Entries (Series)’, click on ‘Series 1’ then click on ‘Remove’. Click ‘OK’.

Excel 2003/Excel Mac 2004 Step 6 • In the 'Title' tab, type the chart title, and the titles for the (X) and (Y) axes. Don't forget to include any units. • Change the gridlines in the 'Gridlines' tab to what looks best. • Remove the legend in the 'Legend' tab. • Click 'Next'. Click 'Finish'. Step 7 Click on the graph, then right-click on one column. Choose 'Format Data Series' → 'Options' and set the 'Gap width' to 0. Click 'OK'.	Excel 2007/Excel Mac 2008 Step 6 In the 'Chart Tools - Design' tab, under 'Chart Layout', choose 'Chart sub-type 8'. Step 7 Double-click on the Chart Title and type in an appropriate title. Repeat for the horizontal and vertical axis titles. Don't forget to include any units. Add borders around the columns. <i>Hint:</i> See step 9. A good title for a histogram is 'Frequency of ...' and then the title of what has been counted.
--	---

Customise and print

Step 8 Add your name so you know which graph is yours when you print it. Do this by:

Excel 2003/Excel Mac 2004 • Clicking on the 'Tools' menu and selecting 'Customise...'. In the 'Toolbars' tab, check the 'Drawing' box and click 'Close'. In the Drawing menu that has just appeared at the bottom of the page, click on the text box icon, then click on an empty space on your graph. Type in your name.	Excel 2007/Excel Mac 2008 • Clicking on a blank corner of the graph window. In the 'Chart Tools', 'Layout' tab click on 'Text box', then click on an empty space on the graph window. Type in your name.
---	--

Step 9 Change the appearance of your graph. First select the graph by clicking on it, then right-click on different parts of the graph. Options are available to customise the background colours, the fonts used, and more.

Step 10 Print Preview can save a lot of printing money and the environment by not wasting paper. It shows what will be printed before it is sent to the printer. Print the graph by:

Excel 2003/Excel Mac 2004 • Clicking on the graph, then click on Print Preview. • If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.	Excel 2007/Excel Mac 2008 • Clicking on the graph, then click on 'Print Preview'. 'Print Preview' is located in the top left Office button under 'Print'. If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.
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Step 11 Save and close the file. Paste the graph into your book.

3 Use the *Hint* in Step 1 to quickly fill in columns in an Excel spreadsheet with these numbers.

- a** 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100

c 70, 71, 72, 73, 74, 75, 76, 77
- b** 0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30

d 0, 0.1, 0.2, 0.3, 0.4, 0.5

4 Write clear, step-by-step instructions for how to change the colours of the columns on a histogram.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Microsoft Excel is a program used to organise data, perform calculations and create graphs.

Excel is made up of worksheets that look like large tables. Each column has a unique letter, and each row has a unique number. Each rectangle is called a cell and each cell has unique co-ordinates based on its column and row.

APPLE MAC ALERT!

In the steps below, replace:

- ‘right-click’ with ‘hold down the Ctrl key and click’
- ‘hold down the Ctrl key’ with ‘hold down the apple/command key’.

- 1
- On the sample Excel worksheet shown:
- a
- draw a cross in cell D1.
- b
- identify what is typed in the formula bar.

- 2
- Follow the steps to create a line graph of ‘Distance from Earth of the Voyager 2 space probe’ using the data shown in the sample Excel worksheet.

Hints:

- Never leave cells within the data table blank. If text is too long for a cell, move the mouse over the border between the column labels A and B so this symbol appears:  Click and hold the left mouse button and drag the column wider. Rows can also be made taller this way.
- To edit the data typed into a cell, first double-click on the cell. With only one click, everything in the cell will be overwritten.

Sample Excel worksheet

B3					6.30e8	← Formula bar
	A	B	C	D		
1	Months after launch	Distance from Earth (km)				
2	0	0				
3	23	6.30e8				
4	48	1.276E+09				
5	101	2.701E+09				
6	144	4.351E+09				
7						

Create a data table

- Step 1
- Enter the data in the sample Excel worksheet into your own Excel worksheet, exactly as shown.
- Hint:
- Enter the data in column B as shown in cell B3. Excel automatically converts 6.30e8 to 6.30E+08. The ‘E+’ in 6.30E+8 represents the ‘× 10’ in 6.30×10^8 .
- Step 2
- Excel often rounds data to two decimal places. To change the number of decimal places, highlight the cells B3 to B6, then:

Excel 2003/Excel Mac 2004 • Click on ‘Format’, then ‘Cells’ and in the ‘Number’ tab choose ‘Scientific’ and adjust the number of decimal places to 3. Click ‘OK’.	Excel 2007/Excel Mac 2008 • In the ‘Home’ tab, ‘Number’ section, click on the drop-down menu and choose ‘Scientific’. Click ‘Increase decimal’ to adjust the number of decimal places to 3.
--	--

- Step 3
- Double-click on the ‘Sheet1’ tab at the bottom of the screen and rename it ‘Line graph’. Save the Excel file by clicking on the disk icon. Name the file ‘Line graph’, then choose a folder to save it in and click ‘OK’.

Don’t forget to save your work regularly.

Create the graph

- Step 4
- Highlight cells A2 to B6 by clicking on A2 and, holding down the mouse button, dragging to cell B6.

Excel 2003/Excel Mac 2004 Step 5 Click on the Chart Wizard and choose ‘XY (Scatter)’ chart type. Choose ‘chart sub-type 4’. Click ‘Next’, then ‘Next’ again.	Excel 2007/Excel Mac 2008 Step 5 Click on the ‘Insert’ tab. Click on ‘Scatter’ chart type and choose ‘chart sub-type 4’.
---	---

Excel 2003/Excel Mac 2004

Step 6

- In the 'Title' tab, type the chart title, and the titles for the 'X' and 'Y' axes. Don't forget to include units if necessary.
- Click on the 'Gridlines' tab to change the gridlines to what looks best.
- Remove the legend in the 'Legend' tab.
- Click 'Next'. Click 'Finish'.

Excel 2007/Excel Mac 2008

Step 6

- In the 'Chart Tools', 'Layout' tab, click on 'Axis Titles' and add a Primary Horizontal Axis Title. Also add a rotated Primary Vertical Axis Title. Double-click on each and type appropriate titles. Don't forget to include any units.
- Click on 'Chart Title' and choose 'Above Chart'. Double-click on the newly created chart title and type in a title.
- • Click on 'Legend' and choose 'None'.

Step 7 Excel sometimes graphs scientific notation numbers as normal numbers. To change the y-axis numbers to scientific numbers, right-click on the y-axis and choose 'Format axis...' then in the 'Number' tab, choose 'Scientific' and adjust the decimal places. Click 'Close' or 'OK'.

Customise and print

Step 8 Add your name so you know which graph is yours when you print it. Do this by:

A good title for a line graph is a description of what is shown in the graph.

Excel 2003/Excel Mac 2004

- Clicking on the 'Tools' menu and selecting 'Customise...'. In the 'Toolbars' tab, check the 'Drawing' box and click 'Close'. In the Drawing menu that has just appeared at the bottom of the page, click on the text box icon, then click on an empty space on your graph. Type in your name.

Excel 2007/Excel Mac 2008

- Clicking on a blank corner of the graph window. In the 'Chart Tools', 'Layout' tab click on 'Text box', then click on an empty space on the graph window. Type in your name.

Step 9 Change the appearance of your graph. First select the graph by clicking on it, then right-click on different parts of the graph. Options are available to customise the background colours, the fonts used, and more.

Step 10 Print Preview can save a lot of printing money and the environment by not wasting paper. It shows what will be printed before it is sent to the printer. Print the graph by:

Excel 2003/Excel Mac 2004

- Clicking on the graph, then click on Print Preview.
- If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.

Excel 2007/Excel Mac 2008

- Clicking on the graph, then click on 'Print Preview'. 'Print Preview' is located in the top left Office button under 'Print'. If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.

Step 11 Save and close the file. Paste the graph into your book.

3 Identify the trend that is shown in the graph.

4 a In Step 2, the Number Format was changed to Scientific, but there are many other Number Formats. Enter the data shown into your own worksheet. Then change the data to different Number Formats.

b Choose one Number Format and describe its effect on the data.

Number Format: _____

Effect on data: _____

K14		
	A	B
1	23.45	
2	65	
3	70	
4	64.9803	
5	0.0012	
6	78000	



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Microsoft Excel is a program used to organise data, perform calculations and create graphs.

Excel is made up of worksheets that look like large tables. Each column has a unique letter, and each row has a unique number. Each rectangle is called a cell and each cell has unique co-ordinates based on its column and row.

APPLE MAC ALERT!

In the steps below, replace:

- ‘right-click’ with ‘hold down the Ctrl key and click’
- ‘hold down the Ctrl key’ with ‘hold down the apple/command key’.

1 Follow the steps to create a scatter graph of ‘How reaction time changes with temperature’ using the data shown in the sample Excel worksheet.

- Hints:
- Never leave cells within the data table blank. If text is too long for a cell, move the mouse over the border between the column labels A and B so this symbol appears: 
 - Click and hold the left mouse button and drag the column wider. Rows can also be made taller this way.
 - To edit the data typed into a cell, first double-click on the cell. With only one click, everything in the cell will be overwritten.

Sample Excel worksheet

	A	B	C	D	E
1	Time (s)				
2	Temperature (°C)	Trial 1	Trial 2	Trial 3	Average
3	10	55	54	56	
4	20	49	49	49	
5	30	43	46	43	
6	40	40	38	38	
7	50	35	37	34	
8	60	31	31	28	
9					

Don't forget to save your work regularly.

Create a data table

Step 1 Enter the data in the sample Excel worksheet into your own Excel worksheet, exactly as shown.

Excel 2003/Excel Mac 2004

- If the text is too long for a cell, click on the ‘Format’ menu, choose ‘Cells’ and in the ‘Alignment’ tab check the ‘Wrap text’ box.
- To superscript °C, highlight the ‘o’ then in ‘Format’ choose ‘Cells’, tick the superscript box and click ‘OK’.
- To make the ‘Time (s)’ heading stretch across all the ‘Trial’ and ‘Average’ columns, highlight cells B1 to E1 then click on ‘Merge & Center’.

Excel 2007/Excel Mac 2008

- If text is too long for a cell, click on ‘Wrap text’ in the ‘Home’ tab under ‘Alignment’.
- To superscript °C, highlight the ‘o’, then in the ‘Home’ tab, click on the small icon in the bottom right hand corner of the ‘Font’ section and tick the superscript box. Click ‘OK’.
- To make the ‘Time (s)’ heading stretch across all the ‘Trial’ and ‘Average’ columns, highlight cells B1 to E1 then click on ‘Merge & Center’.

Step 2 Excel can be used to calculate the data for the ‘Averages’ column. Click on the cell E3. Then:

Excel 2003/Excel Mac 2004

- Click on ‘Insert’, then ‘Function...’. Click on ‘AVERAGE’ and click ‘OK’. If AVERAGE is not in the ‘Most Commonly Used’ Category, find it in the ‘All’ category.

Excel 2007/Excel Mac 2008

- In the ‘Formulas’ tab, click ‘Insert function’. Choose ‘AVERAGE’ and click ‘OK’. If AVERAGE is not in the ‘Most Commonly Used’ Category, find it in the ‘All’ category.

Highlight the cells you want to find the average of, B3 to D3. This range should appear as Number1 B3:D3. Click ‘OK’. Copy the average calculation for cells E4 to E8 by clicking on cell E3, then moving the mouse over the bottom right corner of the cell until it forms a small black cross. Click on the cross and drag it downwards. Let go of the mouse button.

Step 3 Double-click on the ‘Sheet1’ tab at the bottom of the screen and rename it ‘Scatter graph’. Save the Excel file by clicking on the disk icon. Name the file ‘Scatter graph’, then choose a folder to save it in and click ‘OK’.

Create the graph

Step 4 For data with averages, only the average should be plotted on the graph, not all the data. Highlight cells A3 to A8 and E3 to E8. To highlight columns that are not next to each other, highlight cells A3 to A8, let go of the mouse button, hold down the Ctrl key, then highlight cells E3 to E8. Let go of the mouse button and the Ctrl key.

Excel 2003/Excel Mac 2004

Step 5 Click on the Chart Wizard and choose 'XY (Scatter)' chart type. Choose 'chart sub-type 1'. Click 'Next'. Click 'Next' again.

Step 6

- In the 'Title' tab, type the chart title, and the titles for the 'X' and 'Y' axes. Don't forget to include units if necessary.
- Click on the 'Gridlines' tab to change the gridlines to what looks best.
- Remove the legend in the 'Legend' tab.
- Click 'Next'. Click 'Finish'.
- The vertical axis scale can be adjusted by right-clicking on the axis scale, clicking on 'Format Axis' and in 'Axis Options' fixing the minimum at 25. Click 'OK'.

A good title for a scatter graph is 'Y-axis title versus X-axis title'.

Excel 2007/Excel Mac 2008

Step 5 Click on the 'Insert' tab, then click on 'Scatter' chart type. Choose 'chart sub-type 1'.

Step 6

- In the 'Chart Tools', 'Layout' tab, click on 'Axis Titles' and add a Primary Horizontal Axis Title. Also add a rotated Primary Vertical Axis Title. Double-click on each and type appropriate titles. Don't forget to include any units.
- Click on 'Chart Title' and choose 'Above Chart'. Double-click on the newly created chart title and type in a title.
- Click on 'Legend' and choose 'None'.
- The vertical axis scale can be adjusted by right-clicking on the axis scale, clicking on 'Format Axis' and in 'Axis Options' fixing the minimum at 25. Click 'Close'.

Step 7 Add a line of best fit. Right-click on one of the plotted points, choose 'Add trendline...', then click on 'Linear' and click 'OK' or 'Close'. If plotted points look like they should have a curved line of best fit rather than a straight line, try different trendlines such as power or polynomial – but if they don't fit well, it is best to print the graph and draw the line by hand.

Customise and print

Step 8 Add your name so you know which graph is yours when you print it. Do this by:

Excel 2003/Excel Mac 2004

- Clicking on the 'Tools' menu and selecting 'Customise...'. In the 'Toolbars' tab, check the 'Drawing' box and click 'Close'. In the Drawing menu that has just appeared at the bottom of the page, click on the text box icon, then click on an empty space on your graph. Type in your name.

Excel 2007/Excel Mac 2008

- Clicking on a blank corner of the graph window. In the 'Chart Tools', 'Layout' tab, click on 'Text box', then click on an empty space on the graph window. Type in your name.

Step 9 Change the appearance of your graph. First select the graph by clicking on it, then right-click on different parts of the graph. Options are available to customise the background colours, the fonts used, and more.

Step 10 Print Preview can save a lot of printing money and the environment by not wasting paper. It shows what will be printed before it is sent to the printer. Print the graph by:

Excel 2003/Excel Mac 2004

- Clicking on the graph, then click on Print Preview.
- If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.

Excel 2007/Excel Mac 2008

- Clicking on the graph, then click on 'Print Preview'. 'Print Preview' is located in the top left Office button under 'Print'. If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.

Step 11 Save and close the file. Paste the graph into your book.

2 Use Excel to calculate the average of these numbers. *Hint:* See Step 2.

a 90.9, 78.8, 92.4, 86.5, 102.3 Average = ____

b 203, 200, 197, 203, 210, 209 Average = ____

3 Add different trendlines to your graph to see how well each one fits.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Microsoft Excel is a program used to organise data, perform calculations and create graphs.

Excel is made up of worksheets that look like large tables. Each column has a unique letter, and each row has a unique number. Each rectangle is called a cell and each cell has unique co-ordinates based on its column and row.

APPLE MAC ALERT!

In the steps below, replace:

- ‘right-click’ with ‘hold down the Ctrl key and click’
- ‘hold down the Ctrl key’ with ‘hold down the apple/command key’.

- 1 On the sample Excel worksheet shown:
- a circle row 9.

b identify what is typed in cell C4. _____

c draw a triangle in cell A7.

d draw a line through column F.
- 2 Follow the steps to create a divided bar graph of ‘Blood groups of different countries’ using the data shown in the sample Excel worksheet.

Sample Excel worksheet

	Q6					
	A	B	C	D	E	F
1	Percentage of each Blood group					
2	Country	O	A	B	AB	
3	Australia	49	38	10	3	
4	Canada	45	43	9	3	
5	Ireland	55	31	11	3	
6	New Zealand	47	38	11	4	
7	UK	44	42	10	4	
8	US	44	42	10	4	
9						

- Hints:
- Never leave cells within the data table blank. If text is too long for a cell, move the mouse over the border between the column labels A and B so this symbol appears:  Click and hold the left mouse button and drag the column wider. Rows can also be made taller this way.
 - To edit the data typed into a cell, first double-click on the cell. With only one click, everything in the cell will be overwritten.

Create a data table

- Step 1 Enter the data in the sample Excel worksheet into your own Excel worksheet, exactly as shown. To make the ‘Percentage of each Blood Group’ heading stretch across all the blood group columns, highlight cells B1 to E1 then:

Bar graphs and divided bar graphs are the only types of graphs to have the first column in a data table plotted on the y-axis.

Excel 2003/Excel Mac 2004

- Click on ‘Merge & Center’. If the text is too long for the merged cell, click on the ‘Format’ menu, choose ‘Cells’ and in the ‘Alignment’ tab check the ‘Wrap text’ box. Adjust the row height if necessary as outlined in the *Hints* above.

Excel 2007/Excel Mac 2008

- In the ‘Home’ tab, click on ‘Merge & Center’. Click on ‘Wrap Text’ and adjust the row height if necessary as outlined in the *Hints* above.

Don’t forget to save your work regularly.

Create the graph

Step 2 Highlight cells A2 to E8 by clicking on A2 and, holding down the mouse button, dragging to cell E8.

Excel 2003/Excel Mac 2004

Step 3 Click on the Chart Wizard and choose Bar sub-type 3. Click on ‘Next’.

Step 4 If the y-axis data is not on the vertical axis, in the ‘Data range’ tab select ‘Series in Columns’. Click ‘Next’.

Excel 2007/Excel Mac 2008

Step 3 Click on the ‘Insert’ tab and click on ‘Bar’. Choose ‘2D-Bar sub-type 3’.

Step 4 If the y-axis data is not on the vertical axis, right-click on a bar and click on ‘Select Data...’. Click ‘Switch Row/Column’ then click ‘OK’.

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Excel 2003/Excel Mac 2004

Step 5

- In the 'Title' tab, type the chart title and the titles for the (X) and (Y) axes.
- Change the gridlines in the 'Gridlines' tab to what looks best.
- Click 'Next'. Click 'Finish'.

A good title for a divided bar graph is a short sentence which includes the x-axis and y-axis titles.

Excel 2007/Excel Mac 2008

Step 5

- In the 'Chart Tools', 'Layout' tab, click on 'Axis Titles' and add a Primary Horizontal Axis Title. Also add a rotated Primary Vertical Axis Title. Double-click on each and type appropriate titles. Don't forget to include any units.
- Click 'Chart Title' and choose 'Above Chart'. Double-click on the newly created chart title and type in a title.

Step 6 Double-click on the 'Sheet1' tab at the bottom of the screen and rename it 'Divided bar graph'. Save the Excel file by clicking on the disk icon. Name the file 'Divided bar graph', then choose a folder to save it in and click 'OK'.

Customise and print

Step 7 Add your name so you know which graph is yours when you print it. Do this by:

Excel 2003/Excel Mac 2004

- Clicking on the 'Tools' menu and selecting 'Customise...'. In the 'Toolbars' tab, check the 'Drawing' box and click 'Close'. In the Drawing menu that has just appeared at the bottom of the page, click on the text box icon, then click on an empty space on your graph. Type in your name.

Excel 2007/Excel Mac 2008

- Clicking on a blank corner of the graph window. In the 'Chart Tools', 'Layout' tab click on 'Text box', then click on an empty space on the graph window. Type in your name.

Step 8 Change the appearance of your graph. First select the graph by clicking on it, then right-click on different parts of the graph. Options are available to customise the background colours, the fonts used, and more.

Step 9 Print Preview can save a lot of printing money and the environment by not wasting paper. It shows what will be printed before it is sent to the printer. Print the graph by:

Excel 2003/Excel Mac 2004

- Clicking on the graph, then click on 'Print Preview'.
- If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.

Excel 2007/Excel Mac 2008

- Clicking on the graph, then click on 'Print Preview'. 'Print Preview' is located in the top left Office button under 'Print'. If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.

Step 10 Save and close the file. Paste the graph into your book.

	L8				
	A	B	C	D	E
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					

- 3** Write the following data into the Excel worksheet shown so that it could be used to construct a divided bar graph.
- Favourite sport of Year 9 students by class
- Sport – cricket, football, netball, rugby, tennis
- Class 9A – 45, 32, 12, 7, 4
- Class 9B – 16, 27, 44, 10, 3
- Class 9C – 22, 6, 37, 2, 33
- Class 9D – 18, 34, 5, 20, 23

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



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1

Follow the steps to create a composite graph of ‘The effect of adding fertiliser on the height of radish seedlings’.

APPLE MAC ALERT!
In the steps below, replace:

- ‘right-click’ with ‘hold down the Ctrl key and click’
- ‘hold down the Ctrl key’ with ‘hold down the apple/command key’.

Sample Excel worksheet

P19													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1		Plants without fertiliser - Height (mm)						Plants with fertiliser - Height (mm)					
2	Week	Plant 1	Plant 2	Plant 3	Plant 4	Plant 5	Average	Plant 1	Plant 2	Plant 3	Plant 4	Plant 5	Average
3	0	11	10	12	12	10		11	13	11	10	10	
4	1	17	15	16	18	20		23	24	20	25	23	
5	2	29	27	30	27	29		38	36	40	37	33	
6	3	42	45	43	47	44		55	53	57	56	50	
7	4	61	66	65	65	63		85	87	82	84	80	
8	5	77	-	80	81	82		101	99	105	108	100	
9	6	95	-	94	94	97		124	118	125	122	122	
10	7	108	-	109	106	110		149	154	155	156	150	
11	8	122	-	118	125	126		177	176	179	173	179	
12	9	144	-	138	135	149		198	201	204	210	207	
13	10	167	-	155	162	164		229	237	233	236	239	

Create a data table

Don't forget to save your work regularly.

Step 1

Enter the data in your own worksheet, as shown above.

- Plant 2 in the ‘Plants without Fertiliser’ table was eaten by a caterpillar in Week 5. Enter this data as a dash then press Enter. Otherwise, Excel might think you are entering a formula.
- To make the Height (mm) title span several columns, highlight the cells the title needs to stretch across then click on ‘Merge & Center’.
- Adding borders around data makes it easier to understand. Highlight all the cells in the table, then:

Excel 2003/Excel Mac 2004 • Click on ‘Format’, then ‘Cells’, then in the ‘Borders’ tab click on ‘Outline’ and ‘Inside’. Click ‘OK’.	Excel 2007/Excel Mac 2008 • In the ‘Font’ section of the ‘Home’ tab, click on the drop-down menu and select ‘All Borders’.
--	---

Excel 2003/Excel Mac 2004 • Selecting a thicker line style in the ‘Format’, ‘Cells’, ‘Borders’ tab.	Excel 2007/Excel Mac 2008 • In the ‘Font’ section of the ‘Home’ tab, clicking on the drop-down menu and selecting ‘Thick Box Border’.
--	--

• Then add a dark border around the ‘Plants with Fertiliser’ cells.

Step 2

Calculate the average heights for each week. Click on cell G3, as it is the cell where you want to put the first average. Then:

Excel 2003/Excel Mac 2004 • Click on ‘Insert’, then ‘Function...’. Click on ‘AVERAGE’ and click ‘OK’. If AVERAGE is not in the Most Commonly Used Category, find it in the All category.	Excel 2007/Excel Mac 2008 • In the ‘Formulas’ tab, click ‘Insert function’. Choose ‘AVERAGE’ and click OK. If AVERAGE is not in the Most Commonly Used Category, find it in the All category.
---	--

• Check that the ‘Number1’ text box includes the range of cells you want to find the average of – ‘B3:F3’. Click ‘OK’. Move the mouse over the bottom right corner of the cell containing the average until it forms a small black cross. Click on the cross and drag it downwards to fill in all the averages below. Let go of the mouse button. Repeat this step to calculate the average heights for the ‘Plants with Fertiliser’ in column M.

Step 3

Print your data tables. Print Preview can save printing money and the environment by not wasting paper and electricity. It shows what will be printed before it is sent to the printer. Print your data tables by:

Excel 2003/Excel Mac 2004 • In 'File', clicking on Print Preview.	Excel 2007/Excel Mac 2008 • Clicking on Print Preview. Print Preview is located in the top left Office button under 'Print'.
--	---

- If the table is spread across two pages, go to 'Page Setup' and change the Orientation to 'Landscape'. Click 'OK'. If the tables now fit on one page, press 'Print'. Otherwise, select 'Fit to 1 page wide'.

Step 4 Double-click on the 'Sheet1' tab at the bottom of the screen and rename it 'Composite graph'. Save the Excel file by clicking on the disk icon. Name the file 'Composite graph', then choose a folder to save it in and click 'OK'.

Create the graph

Step 5 Only the averages should be plotted, not all the data. Highlight cells A3 to A13, let go of the mouse button, hold down the Ctrl key, then highlight the two columns containing averages data – G3 to G13 and M3 to M13. Let go of the mouse button and the Ctrl key.

Excel 2003/Excel Mac 2004 Step 6 Click on the Chart Wizard and choose 'XY Scatter sub-type 4'. Click 'Next'. Step 7 In the 'Series' tab, click on 'Series 1'. Type 'without fertiliser' as the 'Name' and click 'OK'. Repeat this step for 'Series 2', naming it 'with fertiliser'. Click 'Next'. Step 8 • In the 'Title' tab, type the chart title, and the titles for the (X) and (Y) axes. Don't forget to include units. • Change the gridlines in the 'Gridlines' tab to what looks best. • Keep the legend on the right. It is essential that a legend is included so it is clear which line is for which plants. • Click 'Next'. Click 'Finish'.	Excel 2007/Excel Mac 2008 Step 6 In the 'Insert' tab, click on 'Scatter'. Choose 'sub-type 4'. Step 7 • In the 'Chart Tools', 'Layout' tab, click on 'Axis Titles' and add a Primary Horizontal Axis Title. Also add a rotated Primary Vertical Axis Title. Double-click on each and type appropriate titles. Don't forget to include any units. • Click on 'Chart Title' and choose 'Above Chart'. Double-click on the newly created chart title and type in a title. Step 8 It is essential that a legend is included so it is clear which line is for which plants. Right-click on one data point and choose 'Select Data'. Under the 'Legend Entries' section, click on 'Series 1', then 'Edit'. Type 'without fertiliser' as the 'Series Name' and click 'OK'. Repeat this step for 'Series 2', naming it 'with fertiliser'. Click 'OK'.
--	--

A good title will depend on the type of composite graph. For a composite line graph, try '*y-axis title* versus *x-axis title* for *type of data in the legend*'.

Customise and print

Step 9 Add your name so you know which graph is yours when you print it. Do this by:

Excel 2003/Excel Mac 2004 • Clicking on the 'Tools' menu and selecting 'Customise...'. In the 'Toolbars' tab, check the 'Drawing' box and click 'Close'. In the Drawing menu that has just appeared at the bottom of the page, click on the text box icon, then click on an empty space on your graph. Type your name.	Excel 2007/Excel Mac 2008 • Clicking on the corner of the graph window. In the 'Chart Tools', 'Layout' tab, click on text box, then click on an empty space on the graph window. Type your name.
---	---

Step 10 You can change the appearance of your graph. Select your graph first by clicking on it, then try right-clicking on different parts of the graph.

Step 11 Print your graph by:

Excel 2003/Excel Mac 2004 • Clicking on the graph, then click on Print Preview. If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.	Excel 2007/Excel Mac 2008 • Clicking on the graph, then click on 'Print Preview'. 'Print Preview' is located in the top left Office button under 'Print'. If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.
--	--

Step 12 Save and close the file. Paste the table and the graph into your book.

- 2 Under the graph, write a conclusion based on the growth of the radishes as shown in the graph.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Microsoft Excel is a program used to organise data, perform calculations and create graphs.

Excel is made up of worksheets that look like large tables. Each column has a unique letter, and each row has a unique number. Each rectangle is called a cell and each cell has unique co-ordinates based on its column and row.

- 1 Follow the steps to create a composite line/column graph. Use either:
- the sample data shown in the worksheet on 'Average monthly maximum and minimum temperatures and rainfall for Sydney'
 - data on average monthly temperatures and rainfall in your local area, researched from the Internet or supplied by your teacher.

- Meteorological websites:**
- Australia – <http://www.bom.gov.au/climate/averages/>
 - UK – <http://www.metoffice.gov.uk/climate/uk/averages/19712000/index.html>
 - Canada – http://www.climate.weatheroffice.ec.gc.ca/climate_normals/index_e.html
 - Or try doing a Google search on 'climate normals for <your country>'.

Don't forget to save your work regularly.

Create a data table

Step 1 Enter the data in the sample Excel worksheet into your own Excel worksheet, as shown, replacing the numbers if you are using your own data.

- Hint:* Excel can complete a series if the first few values are entered. Type 'Jan, Feb, Mar' into cells A2, A3 and A4 respectively, then highlight them and move the mouse over the bottom right corner of the selection until it forms a small black cross. Click on the cross and drag it downwards until Dec appears. Let go of the mouse button.
- If text is too wide for the cell, put the mouse on the border between the rows so this symbol appears:  Click and hold the left mouse button to drag the row taller.

APPLE MAC ALERT!

In the steps below, replace:

- 'right-click' with 'hold down the Ctrl key and click'
- 'hold down the Ctrl key' with 'hold down the apple/command key'.

Sample Excel worksheet

	A	B	C	D	E
		Average Minimum Temperature (°C)	Average Maximum Temperature (°C)	Average Rainfall (mm)	
1	Month				
2	Jan	18.6	25.9	102.5	
3	Feb	18.8	25.7	117.9	
4	Mar	17.5	24.7	129.9	
5	Apr	14.7	22.4	126.1	
6	May	11.5	19.4	121.5	
7	Jun	9.2	16.9	130.6	
8	Jul	8.0	16.3	97.6	
9	Aug	8.9	17.8	81.9	
10	Sep	11.0	19.9	69.2	
11	Oct	13.5	22.1	77.0	
12	Nov	15.6	23.6	83.6	
13	Dec	17.5	25.2	78.1	
14					

Excel 2003/Excel Mac 2004 If the text is still too long, click on the 'Format' menu, choose 'Cells' and in the 'Alignment' tab check the 'Wrap text' box.	Excel 2007/Excel Mac 2008 If your text is still too long, click on 'Wrap text' in the 'Home' tab under 'Alignment'.
Excel 2003/Excel Mac 2004 To make superscripts and subscripts, highlight the text to be super/subscripted, then go to 'Format' and choose 'Cells', tick the superscript or subscript box and click 'OK'.	Excel 2007/Excel Mac 2008 To make superscripts and subscripts, highlight the text to be super/subscripted, then in the 'Home' tab, click on the small icon in the bottom right hand corner of the 'Font' section, tick the superscript or subscript box and click 'OK'.
- Display the data to one decimal place. To do this, highlight the data, then:	
Excel 2003/Excel Mac 2004 Click on 'Format', then 'Cells' and in the 'Number' tab choose 'Number' and set the number of decimal places to 1. Click 'OK'.	Excel 2007/Excel Mac 2008 In the 'Home' tab, 'Number' section, adjust the number of decimal places by clicking on 'Decrease decimal'.

Step 2 Double-click on the 'Sheet1' tab at the bottom of the screen and rename it 'Composite graph'. Save your Excel file by clicking on the disk icon. Name your file 'Composite graph2', then choose a folder to save it in and click 'OK'.

Creating the graph

Step 3 Highlight cells A2 to D13 by clicking on A2 and, holding down the mouse button, dragging to cell D13.

Excel 2003/Excel Mac 2004 Step 4 Click on the Chart Wizard and choose 'Column' chart type. Choose 'chart sub-type 1'. Click 'Next'. Step 5 In the 'Series' tab, click on 'Series 1' and type 'Ave Min. Temp.' as the 'Name'. Name Series 2 'Ave Max. Temp.' and Series 3 'Rainfall'. Click 'Next'. Step 6 • In the 'Title' tab, type the chart title, and the titles for the 'X' and 'Y' axes. Label the Y axis Temperature (°C); rainfall will be added later. Don't forget to include units. • Click on the 'Gridlines' tab to change the gridlines to what looks best. • Click 'Next'. Click 'Finish'. Step 7 • Right-click on one data point for rainfall and choose 'Format Data Series'. Under the 'Axis' tab, choose 'Secondary Axis' and click 'OK'. • Right-click on a blank area of the graph and choose 'Chart Options...'. In the 'Titles' tab, add a 'Second value Y axis' title of 'Rainfall (mm)'. Step 8 • A graph with two y-axis variables is easier to read if each is a different type of graph. Right-click on an Ave Min. Temp. data point and choose 'Chart Type'. Choose 'Line graph sub-type 4' and click 'OK'. Repeat this step for Ave Max. Temp.	Excel 2007/Excel Mac 2008 Step 4 In the Insert tab click on 'Column' chart type. Choose chart sub-type 1. Step 5 As there are two different y-axis variables, temperature and rainfall, the graph needs two y-axes. Right-click on a rainfall data point and choose 'Format Data Series'. In the 'Series Options' tab, click on 'Plot Series on Secondary Axis'. Click 'Close'. Step 6 A graph with two y-axis variables is easier to read if each variable is represented by a different type of graph. Right-click on an Ave Min. Temp. data point and choose 'Change Series Chart Type'. Choose 'Line graph sub-type 4' and click 'OK'. Repeat this step for Ave Max. Temp. Step 7 It is essential that a legend is included so it is clear which line is which. Right-click on any data point and choose 'Select Data'. Under the 'Legend Entries' section, click on 'Series 1', then 'Edit'. Type 'Ave Min. Temp.' as the 'Series Name' and click 'OK'. Repeat this step to name Series 2 'Ave Max. Temp.' and Series 3 'Rainfall'. Click 'OK'. Step 8 • In the 'Chart Tools', 'Layout' tab, click on 'Axis Titles' and add a Primary Horizontal Axis Title. Also add a rotated Primary Vertical Axis Title. Double-click on each and type appropriate titles. Don't forget to include units. • Click on 'Chart Title' and choose 'Above Chart'. Double-click on the newly created chart title and type in a title.
--	--

Customise and print

Step 9 Add your name so you know which graph is yours when you print it. Do this by:

Excel 2003/Excel Mac 2004 • Clicking on the 'Tools' menu and selecting 'Customise...'. In the 'Toolbars' tab, check the 'Drawing' box and click 'Close'. In the Drawing menu that has just appeared at the bottom of the page, click on the text box icon, then click on an empty space on your graph. Type in your name.	Excel 2007/Excel Mac 2008 • Clicking on the corner of the graph window. In the 'Chart Tools', 'Layout' tab, click on text box, then click on an empty space on the graph window. Type your name.
---	--

Step 10 Print the graph by:

Excel 2003/Excel Mac 2004 • Clicking on the graph, then click on Print Preview. • If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.	Excel 2007/Excel Mac 2008 • Clicking on the graph, then click on 'Print Preview'. 'Print Preview' is located in the top left Office button under 'Print'. If your graph looks correct and takes up the whole page, press 'Print'. Otherwise, click 'Close' and change it.
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Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Step 11 Save and close the file. Paste the graph into your book.

1 Types of data

- a** experiments **b** numbers **c** months **d** half, seconds **e** can't, divided, midday, blue **f** order, arrange **g** data, small **h** size, time
- quantitative discrete – **b**, interval categorical – **e**, quantitative continuous – **d**, nominal categorical – **c**, ordinal categorical – **a**

2 Types of graphs

- b** Pie – Used to show a set of percentages or fractions that have large differences in the size of the proportions. Graph has no *x*-axis or *y*-axis. Bar – Used to show a set of percentages or fractions that have small differences in the size of the proportions. Column – Used to show categorical data with a quantitative continuous *y* axis variable. Histogram – Used when the dependent variable is a count of how often something occurs, also known as a *frequency*. Line – Used to show a *trend* between unrelated variables, usually over time. Scatter – Used to find a *relationship* between an independent and dependent variable.
- pie, column, histogram, bar
- a** pie **b** histogram **c** column **d** scatter
- a** column **b** bar
- 4a – Major planet, 4b – Household energy use
- a** scatter **b** histogram **c** line

3 Drawing graph scales

- Step 1: 10 Step 3: $90 \div 10 = 9$
- a**
-
- a**

b

4 Quick quiz

- a** data **b** quantitative **c** categorical **d** continuous **e** discrete **f** nominal **g** ordinal **h** interval
- a** categorical nominal **b** quantitative discrete **c** categorical interval **d** quantitative continuous
- a** column **b** scatter **c** pie/bar **d** line **e** histogram
- a** bar, pie, column, histogram **b** bar
-

5 Pie graphs

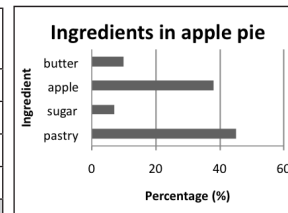
- Percentage of each method of travel to school

Method of travel	Percentage of students (%)	Angle (°)
Bus	42	151
Walk	8	29
Car	30	108
Train	17	61
Bike	3	11

6 Bar graphs

- a** 25% **b** 25%

Ingredient	Mass (g)	Percentage (%)
pastry	95	45
sugar	15	7
apple	80	38
butter	20	10
Total	210	100



7 Column graphs and histograms

- Each axis has a title, graph has a title, equal spaces between the columns, equally spaced scale on *y*-axis, first column of data on the horizontal axis, second column of data on the vertical axis, vertical axis title has units.
- a** Spaces between columns, 90–94 age group incorrectly plotted, no graph title, no horizontal axis title, no units (00's of people) for vertical axis. **b** 65–69 **c** Very few people are diagnosed with lung cancer at younger ages **d** 25–29 and 30–34
- b** 2 **c** 6 **d** 82

8 Line and scatter graphs

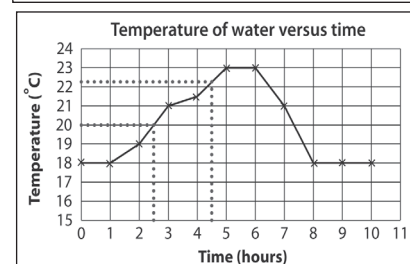
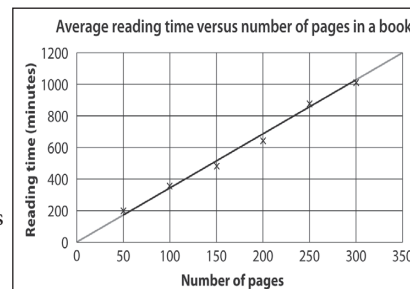
- A** graph title **B** units for axis title **C** $\times$ for each plotted point **D** ruled line joining each point **E** evenly spaced scales **F** ends of axes have arrowheads **G** scale starts at 0 **H** graph takes up maximum space **I** axis title
- a** 10 minutes **b** After 5 minutes **c** 0–1 minutes, 5–6 minutes, 9–10 minutes **d** beats per minute
- b** Force increases as mass increases. Force is linearly proportional to mass.
- a** Time versus age **b** Average height versus mass

9 Lines of best fit

- straight, ruler, hand, first, last, not, equal, exactly
- a** Incorrect, line does not extend just beyond first and last points **b** Incorrect, not an equal number of points on both sides of the line **c** Correct **d** Incorrect, line forced to pass through (0, 0)
-
- A mistake was made taking the measurement or a calculation was done incorrectly or the measuring device stopped working properly or a variable was not controlled.

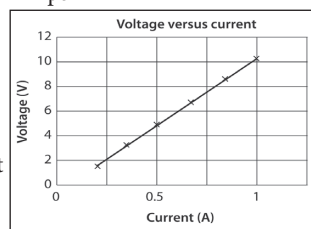
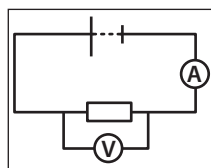
10 Extrapolating and interpolating

- The assumption made is that the relationship remains the same beyond the plotted points.
- Because line and scatter graphs involve continuous data, values between the points are meaningful. They are the only graphs that have lines that are able to be extended.
- Interpolating is more likely to be accurate than extrapolating because the points either side have been measured and are known, whereas for extrapolating only one side of the value has been measured, the other side is unknown.
- 18°C
- No, extrapolating because it extends beyond the plotted points



11 Linear relationships

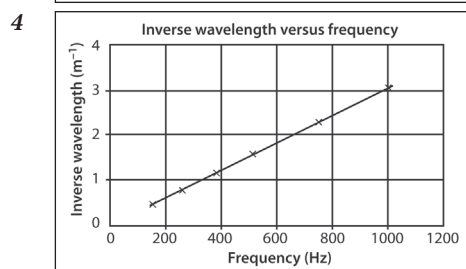
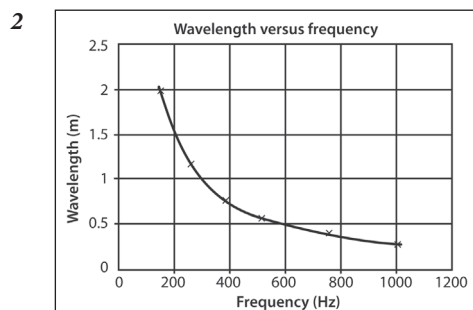
- 2 (A) stands for Amps or Amperes, (V) stands for Volts.
- 4 $y = mx$
- 5 $V = RI$ where V = Voltage measured in Volts (V), I = current measured in Amps (A), R = Resistance measured in Ohms, Ω .
- 6 **d** Approximately 10Ω if the sample data is used.
- 7 Yes
- 8 Will vary according to line of best fit, but approximately: **a** 10V
b 1.5A



12 Non-linear relationships

- 1

Frequency (Hz)	Wavelength (m)	1/wavelength (m^{-1})
150	2.2	0.45
256	1.29	0.78
384	0.86	1.16
512	0.64	1.56
750	0.44	2.27
1000	0.33	3.03



- 7 $\lambda = v/f$, where v = velocity; f = frequency measured in Hertz, Hz; λ = wavelength measured in metres, m
- 8 Approximately 340 m/s, if the sample data is used

- 9

Frequency (Hz)	Wavelength (m)
1 230	Approximately 0.276
Approximately 9 444	0.036
96.1×10^6	Approximately 3.54×10^{-6}

14 Divided bar graphs

- 2

Activity	Omar			Patrick		
	Time (h)	Percent.	Length (cm)	Time (h)	Percent.	Length (cm)
Sleeping	8	33	3.3	8.5	35	3.5
School	7	29	2.9	7	29	2.9
Homework	1	4	0.4	2	8	0.8
Watching TV/Internet	5	21	2.1	3	13	1.3
Other	3	13	1.3	3.5	15	1.5
Total	24	100	10	24	100	10

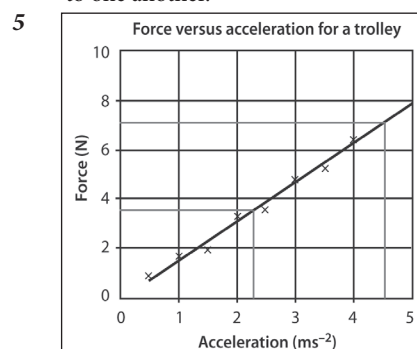
15 Composite line and bar graphs

- 3

Country	Year	Gender	
		Male	Female
Australia	2005	36	32
	2015	40	35
Canada	2005	33	29
	2015	35	31

16 Quick quiz

- 1 Numbers don't increase by equal increments, numbers not marked on major gridlines.
- 2 Distance versus mass
- 3 Both column graphs and line graphs have a horizontal and vertical axes, a graph title and titles for the axes, units are included in the axis title, scales with equal increments. Column graphs have columns whereas line graphs have plotted points joined together, line graphs use continuous data on both axes, however column graphs have categorical data on the x-axis.
- 4 Pie graph, because it involves percentages that are largely different to one another.

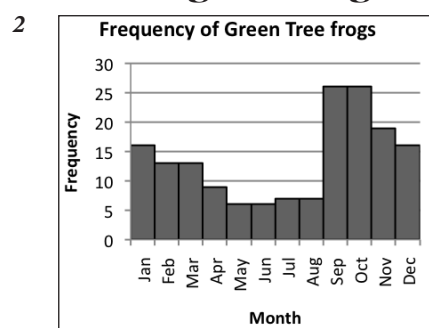


- b** Approximately 4.5 ms^{-2}
c Extrapolating
d Approximately 3.5N
e Interpolating
f Approximately 1.4 kg
g $F = ma$

18 Excelling at bar graphs

- 2 **a** \$B\$7 **b** A2 **3** SUM = 383 **4 a** 11 **b** 9

20 Excelling at histograms



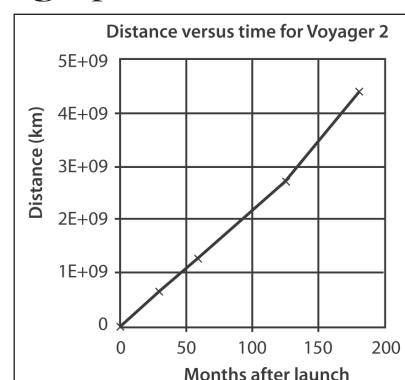
21 Excelling at line graphs

- 1

	A	B
1	Months after launch	Distance from Earth (km)
2	0	0
3	23	6.30E8
4	48	1.276E+09
5	101	2.701E+09
6	144	4.351E+09

Formula bar = 6.30E8
 B3 is the active, or "clicked on", cell – it can have information typed into it.

- 3 Voyager 2 is getting further away, Voyager 2 is speeding up.



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